

Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

A proof of the remaining strict-growth conjecture
and a two-term large-bound asymptotic

Research report prepared for Vladimir Reshetnikov

September 19, 2026

Abstract

Let $a_n^{(m)}$ count permutations $\pi \in S_n$ avoiding the pattern 132 and satisfying $|\pi_{i+1} - \pi_i| \leq m$. Nadler conjectured that their exponential growth constants increase strictly from 1 to 4. Mayama and Akita established a finite-state enumeration, the existence of the growth constants, non-strict monotonicity, and convergence to 4. We prove the missing strict inequality for every adjacency bound. The argument embeds each of the two irreducible cyclic state components at bound m into a proper principal submatrix of its counterpart at bound $m + 1$ by increasing both finite endpoint thresholds by one. Perron–Frobenius comparison then gives strict growth in *each* component, independently of which component dominates.

A separate counting argument gives

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + O(1)}{m}.$$

It compares the full class with a scalar Catalan majorant and an injectively encoded family of variable-length skew blocks. A truncated-series root lemma also bounds the possible constant term. Complete proofs, exact generating functions for $m \leq 4$, integer sequences, and independently checked rational Perron certificates accompany the report.

Addition of 28 September 2026 (batch 38). Part II, from a later manuscript, proves $\alpha_m < \beta_m$ for the root β_m^{-1} of $\sum_{k=1}^m C_{k-1}x^k = 1$ and $\beta_m - \alpha_m = O((\log m)^2/m^2)$; hence the constant term converges, $m(4 - \alpha_m) - 2 \log m - 4 \log \log m \rightarrow 2 \log \pi$, and the growth deficit has an expansion to every inverse-logarithmic order.

Scope and attribution. The finite-state system and fixed- m spectral description are established work, not claimed as new here. The contributions developed in this report are the all- m strict comparison and the refined asymptotic bounds. A targeted literature check found no later resolution of the strict inequality in the sources inspected. This is a research proof with reproducible checks, not a claim of external peer review or an exhaustive novelty certification.

Contents

Part I. Strict growth and the two-term asymptotic	5
1 The problem and the results	5
1.1 Which part was still open?	5
2 The exact endpoint recursion	6
2.1 Decomposition at the maximum	6
2.2 Cumulative endpoint thresholds	6
3 Cyclic components and growth constants	7
3.1 The two relevant components	7
3.2 The strict matrix comparison used in the proof	8
4 A shift of endpoint thresholds proves strict increase	9
5 Scalar upper bounds and injective skew-block lower bounds	10
5.1 A Catalan majorant	10
5.2 Variable-length blocks with recoverable boundaries	11
5.3 Exact block weights and their Catalan tail	12
6 A truncated-series root lemma	13
7 The refined asymptotic and its constant window	14
8 Exact computations and reproducible certificates	15
8.1 Small sequences and rational generating functions	16
8.2 Certified spectral intervals	16
8.3 Large- m scalar comparisons	17
8.4 What was actually checked	18
9 What is resolved, and what remains	19
Part II. The exact constant: an endpoint renewal comparison	20
10 Provenance, conventions and the answer	20
10.1 One manuscript, one addition	20
10.2 Notation of Part II	20
10.3 The question and the answer	22
10.4 What is inherited and what is added	23
11 The finite interface	23
12 The endpoint renewal comparison	24

12.1	A renewal system with a reset	24
12.2	A positive vector for the true two-endpoint component	25
13	A uniform lower approximation from recoverable blocks	25
13.1	Block and component lower bounds	25
13.2	First-entry counts and a uniform tail	26
14	The truncated Catalan root at all orders	27
14.1	A uniform endpoint expansion	27
15	A logarithmically growing cutoff gives a sharp corridor	28
16	Inverting the scalar expansion to every logarithmic order	30
16.1	Finite asymptotic equations, not a convergent series	30
16.2	A triangular recursion for all coefficients	30
16.3	Transfer to the true growth constant	32
16.4	Two different error scales	32
17	Exact checks and numerical evidence	32
17.1	Exact finite checks	32
17.2	Certified finite corridors	33
17.3	Floating-point component and large-bound calculations	33
18	A complete small example and a useful non-implication	36
19	Further research questions and proposed approaches	36
19.1	The first term beyond scalar universality	36
19.2	The eventual ordering of the two components	36
19.3	The geometry of near-critical Perron vectors	37
19.4	Poles, amplitudes, and a uniform spectral gap	37
19.5	Joint growth of length and adjacency bound	37
19.6	Renewal comparisons for other constrained tree classes	37
19.7	Weighted statistics and multivariate perturbations	37
19.8	State minimization versus a spectral majorant	37
19.9	Certified computation and formalization	38
20	Proof boundaries, reproducibility and provenance	38
20.1	What the manuscript has established	38
20.2	Reproducing Part II's computations	38
20.3	Points checked in the written argument	39
20.4	Repository provenance and novelty boundary	39
20.5	Where the merge had to choose	40
	Appendices of Part I	41

A	Exact small-bound algebra	41
B	Reproduction and certificate formats	41
C	Source status and proof audit	42

1 The problem and the results

A permutation $\pi \in S_n$ avoids 132 when there are no indices $i < j < k$ for which $\pi_i < \pi_k < \pi_j$. Fix an integer $m \geq 1$ and put

$$\mathcal{A}_n^{(m)} = \{\pi \in \text{Av}_n(132) : |\pi_{i+1} - \pi_i| \leq m \text{ for } 1 \leq i < n\}, \quad a_n^{(m)} = |\mathcal{A}_n^{(m)}|.$$

We set $a_0^{(m)} = 1$ and use the ordinary generating function

$$A_m(x) = \sum_{n \geq 0} a_n^{(m)} x^n.$$

The growth constant is

$$\alpha_m = \lim_{n \rightarrow \infty} (a_n^{(m)})^{1/n}. \quad (1)$$

The order of limits matters: $n \rightarrow \infty$ is taken for each fixed m , and only afterwards do we let $m \rightarrow \infty$. The trivial identity $a_n^{(m)} = C_n$ for $m \geq n - 1$ does not by itself settle this iterated limit. Here $C_n = \binom{2n}{n}/(n+1)$ is the n th Catalan number.

1.1 Which part was still open?

Nadler's Conjecture 2 in Section 5.2 of [1] asks for the existence of (1), strict increase $1 = \alpha_1 < \alpha_2 < \dots < 4$, and limit 4. Mayama and Akita subsequently proved rationality for all m , existence of the limit, non-strict increase, and $4 - \alpha_m = O(\log m/m)$ [2, Theorems 2.4, 4.12, and 4.14]. Their monotonicity theorem states $\alpha_m \leq \alpha_{m+1}$, not strict inequality.

Merely finding additional permutations at bound $m+1$ is insufficient: sequences 2^n and $2^n + 1$ differ strictly at every index but have identical exponential growth. The needed argument must show that the extra freedom changes the exponential rate.

Theorem 1.1 (Strict increase). *For every integer $m \geq 1$,*

$$\alpha_{m+1} > \alpha_m.$$

In the two-component spectral representation described below, both component rates increase strictly: for every $m \geq 2$,

$$\lambda_U(m+1) > \lambda_U(m), \quad \lambda_V(m+1) > \lambda_V(m).$$

Consequently Nadler's growth-constant conjecture holds in full.

Theorem 1.2 (Two-term approach to the Catalan limit). *As $m \rightarrow \infty$ through integers,*

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + O(1)}{m}. \quad (2)$$

In particular, $4 - \alpha_m \sim 2 \log m/m$. All logarithms in this report are natural logarithms.

Theorem 1.3 (A window for the next constant). *Define $E_m = m(4 - \alpha_m) - 2 \log m - 4 \log \log m$ for $m \geq 2$. Then*

$$2 \log \pi - 4 \log 2 \leq \liminf_{m \rightarrow \infty} E_m \leq \limsup_{m \rightarrow \infty} E_m \leq 2 \log \pi. \quad (3)$$

No convergence of E_m is asserted.

[Added 28 September 2026, batch 38: the last sentence describes the report at its pinned state 6c9175a54. Part II proves that E_m converges to the upper endpoint, $E_m \rightarrow 2 \log \pi$ (theorem 10.2 and corollary 15.3).]

The strict comparison uses a finite-state embedding. The asymptotic results use scalar comparison series and do not require deciding which finite-state component has the larger rate. The distinction is important: the all- m comparison of $\lambda_U(m)$ with $\lambda_V(m)$ is a separate question.

2 The exact endpoint recursion

This section develops the endpoint representation of Mayama and Akita [2, Section 2], including the counting details needed later. It is included to make the subsequent proofs independently readable.

2.1 Decomposition at the maximum

If $\pi = L n R$ avoids 132, every entry of L exceeds every entry of R : otherwise an entry of L , then n , then an entry of R forms 132. Conversely, if this separation holds and both standardized sides avoid 132, their concatenation with n avoids 132.

Suppose R is nonempty and the maximum is at position k . Then L has length $k - 1$, its values are $n - k + 1, \dots, n - 1$, and R has values $1, \dots, n - k$. The jump from n to the first entry of R is at least k . Therefore

$$R \neq \emptyset \implies 1 \leq k \leq m. \quad (4)$$

The alternative $R = \emptyset$ means appending the maximum to an avoider of length $n - 1$.

2.2 Cumulative endpoint thresholds

Let

$$B_m = \{0, 1, \dots, m - 1, \infty\}.$$

For $p, q \in B_m$, let $T_{p,q}^{(m)}(n)$ count the permutations in $\mathcal{A}_n^{(m)}$ satisfying

$$n - \pi_1 \leq p, \quad n - \pi_n \leq q.$$

A threshold ∞ imposes no restriction. Define

$$F_{p,q}^{(m)}(x) = \sum_{n \geq 1} T_{p,q}^{(m)}(n) x^n.$$

A negative threshold gives the zero series, and $\infty - k = \infty$. We suppress the superscript (m) where it is fixed.

For $k \geq 2$ define the nonnegative integer

$$c_{k,p} = |\{\sigma \in \text{Av}_{k-1}(132) : k - \sigma_1 \leq p\}|, \quad c_{1,p} = 1 \quad (5)$$

for nonnegative or infinite p ; for negative p set $c_{k,p} = 0$. These coefficients do not otherwise depend on m . Two properties will be crucial:

$$c_{k,p+1} \geq c_{k,p}, \quad c_{k,\infty} = c_{k-1}. \quad (6)$$

For finite $p \geq 0$ and $k \geq 2$, an explicit formula is

$$c_{k,p} = \sum_{j=1}^{\min(p,k-1)} \frac{j}{k-1} \binom{2k-j-3}{k-2}. \quad (7)$$

A generating-function proof of this formula appears in section 5.3; no numerical fitting is involved.

Proposition 2.1 (Endpoint recurrence). *For all $p, q \in B_m$,*

$$F_{p,q} = x + \sum_{k=1}^m c_{k,p} x^k F_{m-k,q-k} + x F_{p-1,m-1}. \quad (8)$$

Moreover $A_m(x) = 1 + F_{\infty,\infty}(x)$.

Proof. The singleton permutation contributes x to every nonnegative threshold state. For length at least two, first suppose $R \neq \emptyset$ in the maximum decomposition. By (4), $k \leq m$. The left block has $k - 1$ values in an interval of that length, so its internal jumps, and its jump to n , are automatically at most m . Its first-entry restriction gives $c_{k,p}$ choices; for $k = 1$ the first entry is n and the left block is empty.

The right block has length $n - k$. Writing its first and last entries as R_1 and R_{n-k} , the two remaining restrictions are exactly

$$(n - k) - R_1 \leq m - k, \quad (n - k) - R_{n-k} \leq q - k.$$

The first inequality is the jump bound from n to R_1 ; the second is the last-entry threshold for the whole permutation. This gives $c_{k,p} x^k F_{m-k, q-k}$.

If $R = \emptyset$, remove the final maximum. The first deficiency increases by one when the maximum is appended, giving threshold $p - 1$ for the smaller permutation. Its last deficiency must be at most $m - 1$ to permit the last jump. The final entry of the whole permutation is n , so any nonnegative q is satisfied. This is $x F_{p-1, m-1}$. The cases are disjoint and reversible. $\square$

Let $W_m(x)$ have rows and columns indexed by B_m^2 , with

$$(W_m)_{(p,q),(r,s)} = \sum_{k=1}^m c_{k,p} x^k \mathbf{1}_{(r,s)=(m-k, q-k)} + x \mathbf{1}_{(r,s)=(p-1, m-1)}. \quad (9)$$

Invalid negative thresholds are omitted. Rows are the left-hand states of (8); columns are the right-hand states. Thus

$$F = x\mathbf{1} + W_m(x)F, \quad F = x(I - W_m(x))^{-1}\mathbf{1}. \quad (10)$$

Every entry of W_m has zero constant term, so the inverse is well-defined as a formal power series. This also proves rationality and gives an exact algorithm for computing A_m .

3 Cyclic components and growth constants

The following fixed- m structure is established in [2, Proposition 2.7 and Theorem 4.12]. We supply elementary component and transfer-graph arguments to make the spectral input explicit.

3.1 The two relevant components

For $m \geq 2$ put

$$U_m = \{(p, \infty) : 0 \leq p \leq m - 1\}, \quad (11)$$

$$V_m = \{(p, q) : 0 \leq q < p \leq m - 1\} \cup \{(p, m - 1) : 0 \leq p \leq m - 2\}. \quad (12)$$

Their sizes are m and $(m - 1)(m + 2)/2$. Write W_{U_m} and W_{V_m} for the corresponding principal matrices. A directed edge means a nonzero term in a row of (9), directed from its row state to its column state.

Lemma 3.1 (Component structure). *At every real $x > 0$, both $W_{U_m}(x)$ and $W_{V_m}(x)$ are irreducible. The only other cyclic component of the full state graph is the singleton $I_m = (\infty, m - 1)$, whose internal matrix is (x) . Each of these three components is reachable from (∞, ∞) .*

Proof. Within U_m , every state has a $k = 1$ edge to $(m - 1, \infty)$. From that state, the $k = 1, \dots, m$ edges reach every state of U_m , since $c_{k, m-1} > 0$. Hence U_m is strongly connected.

For V_m , call the states with $q < p$ lower states and the states $(p, m - 1)$ top states. Every lower state has an append edge to a top state. Every top state has a $k = 1$ edge to

$$e = (m - 1, m - 2).$$

It remains to show that e reaches every state. For $m = 2$ this is the immediate two-state cycle. Suppose $m \geq 3$. Appending at e reaches $(m - 2, m - 1)$; its split edges reach all lower states $(p, p - 1)$, $1 \leq p \leq m - 1$. Their append edges reach every top state. Inductively, suppose all lower states of gap at most $d - 1$ are reachable, where $2 \leq d \leq m - 1$. The state $(m - 1, m - d)$ is then reachable. Its split edge with $k = m - p$ reaches $(p, p - d)$ for every $d \leq p \leq m - 1$. Thus every lower gap is reached.

For finite q , every valid split edge ends at a lower state, because $(m - k) - (q - k) = m - q > 0$. Every valid append edge with finite p ends at a top state. Therefore a finite-finite state outside V_m cannot be on a cycle. A state (∞, q) with finite q has append target I_m and split targets in V_m ; among these states only I_m has a self-loop. No edge returns to (∞, ∞) . Edges with second coordinate ∞ stay in U_m or leave it by an append edge, and cannot return from a finite second coordinate. This exhausts all possibilities.

Finally, the $k = 1$ split from (∞, ∞) reaches U_m , its append edge reaches I_m , and a $k = 1$ split from I_m reaches $e \in V_m$. $\square$

3.2 The strict matrix comparison used in the proof

We use the Perron–Frobenius theorem for finite nonnegative matrices: an irreducible matrix has positive left and right eigenvectors at its spectral radius; a primitive matrix has powers asymptotic to a positive rank-one matrix times the corresponding power of that radius.

Lemma 3.2 (Proper principal comparison). *Let M be an irreducible nonnegative matrix and let S be a nonempty proper subset of its indices. If $B = M[S, S]$, then $\text{spr}(B) < \text{spr}(M)$. If $0 \leq N \leq B$ entrywise, then $\text{spr}(N) < \text{spr}(M)$ as well.*

Proof. Take a nonnegative nonzero Perron vector v of B and extend it by zeros to a vector w on all indices of M . Then $Mw \geq \text{spr}(B)w$. There is strict inequality in at least one coordinate: the support of w is nonempty and proper, and irreducibility supplies an edge from its complement to that support. If u is a positive left Perron vector of M , multiplication by $u^\top$ gives

$$\text{spr}(M)u^\top w > \text{spr}(B)u^\top w.$$

Since $u^\top w > 0$, the first assertion follows. Entrywise monotonicity of the spectral radius gives $\text{spr}(N) \leq \text{spr}(B)$; for example, this monotonicity follows by comparing powers and then taking matrix-norm roots. $\square$

For $C = U, V$, there is a unique $r_C(m) > 0$ satisfying

$$\text{spr}(W_{C_m}(r_C(m))) = 1, \quad \lambda_C(m) = r_C(m)^{-1}. \quad (13)$$

Indeed the spectral radius is continuous, tends to zero as $x \downarrow 0$, and tends to infinity as $x \rightarrow \infty$ because the component has a positive cycle. If $y > x > 0$, every monomial has degree at least one, so $W_{C_m}(y) \geq (y/x)W_{C_m}(x)$. Its positive spectral radius is therefore strictly increasing.

Proposition 3.3 (Fixed-bound growth). *For $m \geq 2$ the limit (1) exists and*

$$\alpha_m = \max\{\lambda_U(m), \lambda_V(m)\}. \quad (14)$$

Also $\lambda_U(m) > 1$.

Proof. The state $(m-1, \infty)$ has an internal self-loop of weight x . At $x = 1$ its singleton principal matrix is (1) , and it is a proper principal matrix of $W_{U_m}(1)$. By lemma 3.2, the latter has spectral radius greater than one. Thus $r_U(m) < 1$ and $\lambda_U(m) > 1$.

To justify the actual limit rather than only a limsup, replace each edge of weight cx^k by a directed path of k unit-length edges with total multiplicity c . Use fresh intermediate vertices for every edge. Let H be the resulting finite nonnegative adjacency matrix, and let b be one on original endpoint states and zero on intermediate states. With o the state (∞, ∞) , (10) gives

$$a_n^{(m)} = e_o^T H^{n-1} b \quad (n \geq 1).$$

For an expanded irreducible component with Perron eigenvalue λ , eliminating the intermediate coordinates of a positive eigenvector gives $z = W_{C_m}(1/\lambda)z$. Its eigenvalue is therefore $\lambda_C(m)$. The expanded singleton component has eigenvalue one. The component classification and $\lambda_U(m) > 1$ show that $\text{spr}(H)$ is the right-hand side of (14).

The expanded U_m component is primitive, since it is strongly connected and has a length-one loop. For $m \geq 3$, the expanded V_m component has closed walks of lengths two and five. The first is

$$e \xrightarrow{1} (m-2, m-1) \xrightarrow{1} e.$$

For the second, split with $k = 2$ on the middle state and then append and split with $k = 1$:

$$e \xrightarrow{1} (m-2, m-1) \xrightarrow{2} (m-2, m-3) \xrightarrow{1} (m-3, m-1) \xrightarrow{1} e.$$

The labels are edge lengths. All coefficients are positive, including when $m = 3$. Their greatest common divisor is one, so this component is primitive. When $m = 2$, V_m is a two-cycle with rate one and is not dominant.

A dominant primitive component is reachable from o , and b is positive on its original states. Fixing an entrance path and then applying the primitive Perron theorem supplies $a_n^{(m)} \geq c \text{spr}(H)^n$ for all sufficiently large n , with $c > 0$. A finite-matrix power bound gives $a_n^{(m)} \leq Cn^D \text{spr}(H)^n$ for some fixed C, D (or, equivalently, an upper bound by $(\text{spr}(H) + \varepsilon)^n$ for each $\varepsilon > 0$). Taking n th roots proves (14) and the existence of the limit. $\square$

4 A shift of endpoint thresholds proves strict increase

The key comparison preserves the split length k , not merely the underlying unweighted graph. Define

$$\phi(p) = \begin{cases} p+1, & p < \infty, \\ \infty, & p = \infty, \end{cases} \quad \Phi(p, q) = (\phi(p), \phi(q)).$$

The next lemma is the essential new observation.

Lemma 4.1 (Coefficientwise shift embedding). *For every $m \geq 2$ and $C = U, V$, the map Φ embeds C_m into a proper subset of C_{m+1} . Under this identification,*

$$W_{C_m}(x) \leq W_{C_{m+1}}(x)[\Phi(C_m), \Phi(C_m)] \quad (15)$$

coefficientwise in every matrix entry.

Proof. A state (p, ∞) of U_m maps to $(p+1, \infty)$ of U_{m+1} ; the state $(0, \infty)$ of the latter is missing. A lower state (p, q) of V_m maps to a lower state $(p+1, q+1)$ of V_{m+1} . A top state $(p, m-1)$ maps to $(p+1, m)$, a top state at the new bound. The new top state $(0, m)$ is missing, so that inclusion is also proper.

Consider a split term at row (p, q) of the old matrix, with column $(m - k, q - k)$ and coefficient $c_{k,p}x^k$. At the new row $\Phi(p, q)$, the split with the same k has column

$$(m + 1 - k, \phi(q) - k) = \Phi(m - k, q - k),$$

whenever the old term is valid. Its coefficient is $c_{k,\phi(p)}x^k \geq c_{k,p}x^k$. For $p = \infty$ the coefficients are equal. For a valid append term, the old column $(p - 1, m - 1)$ maps to

$$(\phi(p) - 1, m),$$

which is exactly the new append column, and the weight remains x . Each term is preserved with at least its old multiplicity. Adding terms with the same row and column proves the coefficientwise assertion. $\square$

Proof of theorem 1.1. Fix $m \geq 2$, $C \in \{U, V\}$, and a positive real x . By lemmas 3.1, 3.2 and 4.1,

$$\begin{aligned} \text{spr}(W_{C_m}(x)) &\leq \text{spr}(W_{C_{m+1}}(x)[\Phi(C_m), \Phi(C_m)]) \\ &< \text{spr}(W_{C_{m+1}}(x)). \end{aligned}$$

At $x = r_C(m)$ the first quantity is one. Strict increase of the new spectral radius as a function of x gives $r_C(m + 1) < r_C(m)$, hence $\lambda_C(m + 1) > \lambda_C(m)$.

Choose a component C realizing the maximum at bound m . Then

$$\alpha_{m+1} \geq \lambda_C(m + 1) > \lambda_C(m) = \alpha_m.$$

This works even if the dominant component changes between the two bounds.

For $m = 1$, any adjacent difference is ± 1 , and reversing direction would repeat a value. The only permutations of length $n \geq 2$ are the increasing and decreasing ones, both avoiding 132. Thus $\alpha_1 = 1$. At $m = 2$, in the order $(0, \infty), (1, \infty)$,

$$W_{U_2}(x) = \begin{pmatrix} 0 & x \\ x^2 & x \end{pmatrix}, \quad \det(I - W_{U_2}(x)) = 1 - x - x^3.$$

Its positive root is below one, so $\alpha_2 \geq \lambda_U(2) > 1$. This covers the remaining step. $\square$

Remark 4.2. Strictness comes from a *proper principal extension of an irreducible component*. It does not require finding a strictly larger entry inside the old image. This is useful for the U component, where many coefficients are preserved exactly. Nor is a numerical lower bound on the gap between consecutive rates needed.

5 Scalar upper bounds and injective skew-block lower bounds

We next obtain two scalar comparison series whose root asymptotics have the same first two terms. Write $P \leq Q$ for coefficientwise inequality of formal power series with nonnegative coefficients.

5.1 A Catalan majorant

Define

$$R_m(x) = x + \sum_{k=1}^m C_{k-1}x^k. \tag{16}$$

The separate initial x accounts for appending a maximum, whereas the $k = 1$ term in the sum accounts for a maximum in the first position with a nonempty right block. Thus the coefficient of x in R_m is two.

Proposition 5.1 (Scalar upper bound). *Let $q_m > 0$ be the unique positive root of $R_m(q_m) = 1$. Then*

$$A_m(x) \leq 1 + \frac{x}{1 - R_m(x)}, \quad \alpha_m \leq q_m^{-1} < 4. \quad (17)$$

Proof. The maximum decomposition, with all endpoint restrictions discarded, gives for $n \geq 2$

$$a_n^{(m)} \leq a_{n-1}^{(m)} + \sum_{k=1}^{\min(m, n-1)} C_{k-1} a_{n-k}^{(m)}.$$

The upper endpoint $n - 1$ is essential: the right block in a split must be nonempty. Starting with $a_1^{(m)} = 1$, induction yields the generating-function majorant. Its radius is q_m : the denominator is nonzero for $|x| < q_m$ by positivity, and it vanishes at q_m without numerator cancellation. Cauchy–Hadamard therefore gives $\alpha_m \leq q_m^{-1}$.

Finally,

$$R_m(1/4) < \frac{1}{4} + \sum_{k \geq 1} C_{k-1} 4^{-k} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} < 1.$$

Hence $q_m > 1/4$ and the strict upper bound by four follows. $\square$

5.2 Variable-length blocks with recoverable boundaries

Fix $d \geq 1$ and $m > d$. An admissible block of length k , where $1 \leq k \leq m - d$, is a permutation $\beta \in \text{Av}_k(132)$ such that

$$\beta_k = k, \quad k - \beta_1 \leq d.$$

There are $c_{k,d}$ such blocks: for $k \geq 2$, deleting the last maximum leaves the permutations counted in (5); the singleton contributes $c_{1,d} = 1$.

For a sequence of blocks, assign consecutive value intervals in decreasing order. That is, every entry of an earlier block exceeds every entry of a later block. This operation is their *skew sum*. Let

$$K_{m,d}(x) = \sum_{k=1}^{m-d} c_{k,d} x^k. \quad (18)$$

Proposition 5.2 (Injective block lower bound). *Let $s_{m,d} > 0$ be the unique positive root of $K_{m,d}(s_{m,d}) = 1$. Then*

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x), \quad s_{m,d}^{-1} \leq \alpha_m \leq q_m^{-1}. \quad (19)$$

Proof. A skew sum of 132-avoiding blocks avoids 132. If three indices are not all in one block, their first and last indices lie in different blocks, so the first value exceeds the last; this is incompatible with 132. Within a block, the largest possible jump is $k - 1 < m$.

For a boundary, let the earlier block have length k and let the next block have first-entry deficiency $j \leq d$. The earlier block ends at its maximum. Since the two value intervals are consecutive and decreasing, the boundary jump is exactly $k + j \leq m$. This also covers a singleton next block, where $j = 0$.

It remains to prove injectivity; this is not automatic for variable-length concatenations. A skew cut in a permutation is a proper cut for which every value on the left exceeds every value on the right. A block ending at its maximum has no internal skew cut, because its maximum lies to the right of every proper cut. Every actual boundary between the constructed blocks is a skew cut. Any other proposed cut is internal to a block and is not a skew cut. Consequently the complete set of skew cuts recovers exactly the block boundaries, and standardization recovers each block. Distinct block sequences cannot yield the same permutation.

The empty sequence and all nonempty sequences of admissible blocks are therefore counted by $1 + K + K^2 + \dots = (1 - K)^{-1}$ and inject into the desired class. Its radius is $s_{m,d}$ by the same positivity argument as above, which proves the growth lower bound. $\square$

For $d = 1$ this becomes particularly simple:

$$K_{m,1}(x) = x + \sum_{k=2}^{m-1} C_{k-2} x^k. \quad (20)$$

This one-parameter subfamily already proves theorem 1.2. Allowing arbitrary fixed d sharpens the bounded remainder in theorem 1.3.

5.3 Exact block weights and their Catalan tail

Let $h(k, j)$ count blocks of length $k \geq 2$ ending at their maximum and having first entry $k - j$, where $1 \leq j \leq k - 1$. Put

$$C(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}, \quad C(x) = 1 + xC(x)^2.$$

Lemma 5.3 (First-entry generating function). *For each fixed $j \geq 1$,*

$$\sum_{k \geq j+1} h(k, j) x^k = x^{j+1} C(x)^j, \quad h(k, j) = \frac{j}{k-1} \binom{2k-j-3}{k-2}. \quad (21)$$

Consequently

$$K_{\infty,d}(x) := \sum_{k \geq 1} c_{k,d} x^k = x \sum_{j=0}^d (xC(x))^j. \quad (22)$$

Proof. Delete the final maximum from such a block. Its first entry is $a = k - j$. The $j - 1$ larger entries $a + 1, \dots, k - 1$ must appear in increasing order: a decreasing pair together with the initial a would form 132. The smaller entries lie in the j intervening slots, giving the form

$$a, \tau_0, a + 1, \tau_1, a + 2, \dots, k - 1, \tau_{j-1}.$$

Every value in an earlier small slot must exceed every value in a later small slot. Otherwise those two values and a separating large entry form 132. Each slot must itself avoid 132. Conversely these conditions are sufficient: triples meeting different small slots cannot form 132, the large entries are increasing, and any triple entirely in one small slot is excluded by its own avoidance condition. Once the slot lengths are specified, their value intervals are forced in decreasing order. There are therefore j independent Catalan slots and $j + 1$ fixed entries when the deleted final maximum is restored. This gives the first identity.

For completeness, put $Y = C(x) - 1$, so $Y = x(1 + Y)^2$. For $N \geq 1$, Lagrange inversion gives

$$[x^N] C(x)^j = \frac{j}{N} [u^{N-1}] (1 + u)^{2N+j-1} = \frac{j}{2N+j} \binom{2N+j}{N}.$$

The same final expression is one for $N = 0$. Substituting $N = k - j - 1$ and cancelling factorials gives the displayed formula for $h(k, j)$. Summing over $1 \leq j \leq d$, with the singleton added, proves (22) and (7). $\square$

Since $C(1/4) = 2$, (22) yields

$$B_d := \sum_{k \geq 1} c_{k,d} 4^{-k} = \frac{1}{2} (1 - 2^{-(d+1)}). \quad (23)$$

To obtain the tail constant without singularity-analysis machinery, compare each first-entry count directly with a Catalan number:

$$\frac{h(k, j)}{C_{k-2}} = j \prod_{r=0}^{j-2} \frac{k-2-r}{2k-4-r} \longrightarrow \frac{j}{2^{j-1}} \quad (k \rightarrow \infty). \quad (24)$$

The empty product for $j = 1$ is one. Stirling's formula gives $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$. For fixed d it follows that

$$c_{k,d} 4^{-k} \sim \frac{a_d}{4\sqrt{\pi}} k^{-3/2}, \quad a_d := \sum_{j=1}^d \frac{j}{2^{j+1}} = 1 - \frac{d+2}{2^{d+1}}. \quad (25)$$

Both a_d and B_d are strictly positive for $d \geq 1$, and $B_d < 1$.

6 A truncated-series root lemma

The following elementary estimate is the analytic step responsible for the $\log \log m$ term. It applies to both sides of (19).

Lemma 6.1 (A tilted $k^{-3/2}$ tail). *Let $b_k \geq 0$ satisfy*

$$b_k \sim c k^{-3/2}, \quad c > 0, \quad B := \sum_{k \geq 1} b_k < 1.$$

Fix an integer $h \geq 0$. For all sufficiently large m , let $r_m > 0$ be the unique positive solution of

$$\sum_{k=1}^{m-h} b_k (4r_m)^k = 1.$$

Then $r_m > 1/4$ and

$$m \log(4r_m) = \frac{1}{2} \log m + \log \log m + \log \frac{1-B}{2c} + o(1). \quad (26)$$

Proof. At $r = 1/4$ the truncated sum is at most $B < 1$. For large enough m the polynomial is nonconstant with a positive coefficient, strictly increases on the positive real axis, and tends to infinity. Thus the stated root exists uniquely and exceeds $1/4$.

For a fixed real s , write

$$t = t_m(s) = \frac{1}{2} \log m + \log \log m + s.$$

We first prove the limiting profile

$$\sum_{k=1}^{m-h} b_k e^{tk/m} \longrightarrow B + 2ce^s. \quad (27)$$

The untilted truncated sum tends to B . It remains to evaluate the increment from exponential tilting.

For $k \leq m/2$, use $e^u - 1 \leq ue^u$ for $u \geq 0$ to obtain

$$\begin{aligned} 0 \leq \sum_{k \leq m/2} b_k(e^{tk/m} - 1) &\leq \frac{t}{m} e^{t/2} \sum_{k \leq m/2} kb_k \\ &= O\left(\frac{te^{t/2}}{\sqrt{m}}\right) = O\left(m^{-1/4}(\log m)^{3/2}\right) = o(1). \end{aligned}$$

Here $b_k = O(k^{-3/2})$ implies $\sum_{k \leq M} kb_k = O(\sqrt{M})$.

For $k > m/2$, the untilted tail is $O(m^{-1/2}) = o(1)$. The assumed asymptotic for b_k holds uniformly in this range as $m \rightarrow \infty$. Put $j = m - k$. The tilted upper tail is therefore asymptotic to

$$cm^{-3/2}e^t \sum_{h \leq j < m/2} (1 - j/m)^{-3/2} e^{-tj/m}. \quad (28)$$

The sum in (28) is asymptotic to m/t . To check this precisely, for $j \leq m/\sqrt{t}$ the factor $(1 - j/m)^{-3/2}$ is $1 + O(t^{-1/2})$ uniformly. The geometric sum without that factor is

$$\sum_{j \geq h} e^{-tj/m} = \frac{e^{-th/m}}{1 - e^{-t/m}} \sim \frac{m}{t}.$$

The omitted part $j > m/\sqrt{t}$, with $j < m/2$, is bounded by a constant times $e^{-\sqrt{t}}/(1 - e^{-t/m}) = o(m/t)$, since $(1 - j/m)^{-3/2} \leq 2^{3/2}$. The tail beyond $m/2$ in the geometric sum is negligible by the same estimate. A fixed lower cutoff h has no effect on the asymptotic.

Thus (28) is asymptotic to

$$\frac{ce^t}{\sqrt{m}t} = \frac{ce^s \log m}{\frac{1}{2} \log m + \log \log m + s} \longrightarrow 2ce^s,$$

which proves (27).

Let $s_0 = \log((1 - B)/(2c))$. For each fixed $\varepsilon > 0$, the limiting value in (27) is below one at $s = s_0 - \varepsilon$ and above one at $s = s_0 + \varepsilon$. Monotonicity of the polynomial in t traps $m \log(4r_m)$ between $t_m(s_0 - \varepsilon)$ and $t_m(s_0 + \varepsilon)$ for all sufficiently large m . Since ε is arbitrary, (26) follows. $\square$

Remark 6.2 (Why the second logarithm appears). The effective large- k contribution comes from a boundary layer of width about m/t . Each coefficient near the cutoff is of order $m^{-3/2}$, and the exponential factor is e^t . Its total mass is consequently $e^t/(\sqrt{m}t)$. Requiring a nonzero limiting mass gives $t = \frac{1}{2} \log m + \log t + O(1)$, hence the additional $\log \log m$. The proof above supplies the constants and controls the discarded ranges.

7 The refined asymptotic and its constant window

Put

$$L_m = \frac{1}{2} \log m + \log \log m, \quad t_m^* = m \log \frac{4}{\alpha_m}.$$

We apply lemma 6.1 with two different nonnegative coefficient sequences.

For the upper-growth majorant R_m , write

$$R_m(x) = \sum_{k=1}^m b_k(4x)^k, \quad b_1 = \frac{1}{2}, \quad b_k = C_{k-1}4^{-k} \quad (k \geq 2).$$

Then

$$B = \frac{3}{4}, \quad c = \frac{1}{4\sqrt{\pi}}, \quad h = 0.$$

Therefore

$$m \log(4q_m) = L_m + \log \frac{\sqrt{\pi}}{2} + o(1). \quad (29)$$

For the block lower bound, take $b_k = c_{k,d}4^{-k}$ and $h = d$, holding d fixed as $m \rightarrow \infty$. Equations (23) and (25) give

$$m \log(4s_{m,d}) = L_m + \gamma_d + o(1), \quad \gamma_d = \log \left(\frac{\sqrt{\pi}(1 + 2^{-(d+1)})}{1 - (d+2)2^{-(d+1)}} \right). \quad (30)$$

The $o(1)$ in this formula is for each fixed d ; no uniformity in a growing d is used.

Proof of theorem 1.2. From (19),

$$m \log(4q_m) \leq t_m^* \leq m \log(4s_{m,d}). \quad (31)$$

Already $d = 1$, for which $\gamma_1 = \log(5\sqrt{\pi})$, shows $t_m^* = L_m + O(1)$. Hence $\alpha_m = 4e^{-t_m^*/m}$ and

$$m(4 - \alpha_m) = 4m(1 - e^{-t_m^*/m}) = 4t_m^* + O\left(\frac{(t_m^*)^2}{m}\right) = 4t_m^* + o(1). \quad (32)$$

Since $4L_m = 2 \log m + 4 \log \log m$, the theorem follows. This also independently proves $\alpha_m \rightarrow 4$. $\square$

Proof of theorem 1.3. The lower side of (31), together with (29) and (32), gives

$$\liminf_{m \rightarrow \infty} E_m \geq 4 \log(\sqrt{\pi}/2) = 2 \log \pi - 4 \log 2.$$

For every fixed $d \geq 1$, the upper side gives

$$\limsup_{m \rightarrow \infty} E_m \leq 4\gamma_d.$$

Now let $d \rightarrow \infty$ after taking the limsup in m . Because $\gamma_d \rightarrow \log \sqrt{\pi}$, the upper bound becomes $2 \log \pi$. The middle inequality between liminf and limsup is automatic. $\square$

Remark 7.1 (The constants are not finite- m error bars). The endpoints in (3) are asymptotic bounds, not a claim that every E_m lies in that interval. Numerically the scalar comparisons approach their limits slowly. Even at $m = 10^6$, the best of the four computed fixed- d lower bounds gives a remainder upper bound above $2 \log \pi$. This is compatible with the theorem: a fixed d has limit $4\gamma_d$, and the root expansion also has a slowly vanishing error.

[Added 28 September 2026, batch 38: with the sharper upper bound and the growing cutoff of Part II, the computed corridor for E_m at $m = 10^6$ is about $[3.1495, 3.1503]$, still far above the proved limit $2 \log \pi$; see remark 17.1.]

8 Exact computations and reproducible certificates

The proofs above hold for all m and do not rest on a finite search. The computations serve three different purposes: checking the combinatorial implementation against direct enumeration, providing exact certificates for displayed finite cases, and illustrating the scalar bounds at large m . These evidence levels are kept separate.

Table 1: Initial counts computed from the exact endpoint recurrence. The empty permutation is included at $n = 0$.

n	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$
0	1	1	1	1	1	1
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	2	5	5	5	5	5
4	2	8	14	14	14	14
5	2	12	28	42	42	42
6	2	18	55	95	132	132
7	2	26	108	216	322	429
8	2	37	214	494	799	1101
9	2	53	412	1135	2005	2892
10	2	76	787	2613	5054	7714
11	2	109	1497	5892	12758	20714
12	2	157	2841	13218	32140	55697

8.1 Small sequences and rational generating functions

For example, exact symbolic elimination gives

$$A_1(x) = \frac{1+x^2}{1-x}, \quad A_2(x) = \frac{1-2x+2x^2-x^4-x^6}{(1-x)^2(1-x-x^3)}. \quad (33)$$

Thus, with our convention $a_0^{(2)} = 1$,

$$a_n^{(2)} = 3a_{n-1}^{(2)} - 3a_{n-2}^{(2)} + 2a_{n-3}^{(2)} - 2a_{n-4}^{(2)} + a_{n-5}^{(2)} \quad (n \geq 7).$$

The lower threshold on n comes from the numerator degree, not just the denominator degree. The archive includes exact numerator and denominator coefficient arrays for $m = 1, 2, 3, 4$, and coefficients through $n = 200$ for $m = 1, \dots, 12$.

These rational functions are obtained by exact polynomial-matrix inversion, not by guessing recurrences from the initial terms. More importantly, *the verifier does not need to trust the symbolic solver*. For every $m \leq 4$, the supplied certificate gives a common denominator $D(x)$ and numerator polynomials $N_{p,q}(x)$ for all endpoint states, and verifies exactly that

$$(I - W_m(x))N(x) = xD(x)\mathbf{1}, \quad D(0) = 1. \quad (34)$$

Uniqueness of the formal solution then proves all coefficients, not merely the first 200 checked by an additional test. Integer cross multiplication also verifies the reported reduced expression for $1 + F_{\infty, \infty}$.

8.2 Certified spectral intervals

For each $m = 2, \dots, 20$ and each component, a numerical eigensolver proposes an approximate positive Perron vector and an approximate radius. These proposals are converted to positive integer vectors and rational endpoints. They are accepted only when integer arithmetic verifies

$$W_{C_m}(\ell)v < v, \quad W_{C_m}(u)w > w \quad \text{coordinatewise}, \quad v, w > 0. \quad (35)$$

Positive left Perron vectors show that these imply $\text{spr}(W_{C_m}(\ell)) < 1 < \text{spr}(W_{C_m}(u))$. Consequently

$$\ell < r_C(m) < u, \quad \frac{1}{u} < \lambda_C(m) < \frac{1}{\ell}.$$

The verifier uses no floating-point operations for these assertions. If $\ell = p/D$, an edge of degree k contributes the integer factor $p^k D^{m-k}$ after clearing the common denominator D^m .

For example, the $m = 2$, U certificate gives

$$\frac{6823278035}{10^{10}} < r_U(2) < \frac{6823278042}{10^{10}}.$$

All 38 component certificates pass. Their disjoint intervals additionally certify the component comparison for every $m = 2, \dots, 20$: U dominates for $m = 2, 3, 4$ and V for $m = 5, \dots, 20$. This finite conclusion is not extrapolated into an all- m theorem.

Table 2: Component growth rates, rounded to six decimals. The component ordering is certified by rational intervals; displayed decimals are rounded numerical values.

m	$\lambda_U(m)$	$\lambda_V(m)$	α_m	Dominant
2	1.465571	1.000000	1.465571	U
3	1.826516	1.691403	1.826516	U
4	2.100456	2.091477	2.100456	U
5	2.312119	2.351509	2.351509	V
6	2.479693	2.535762	2.535762	V
8	2.727583	2.785681	2.785681	V
10	2.902175	2.952689	2.952689	V
12	3.032016	3.074770	3.074770	V
15	3.174821	3.208325	3.208325	V
20	3.333400	3.356796	3.356796	V

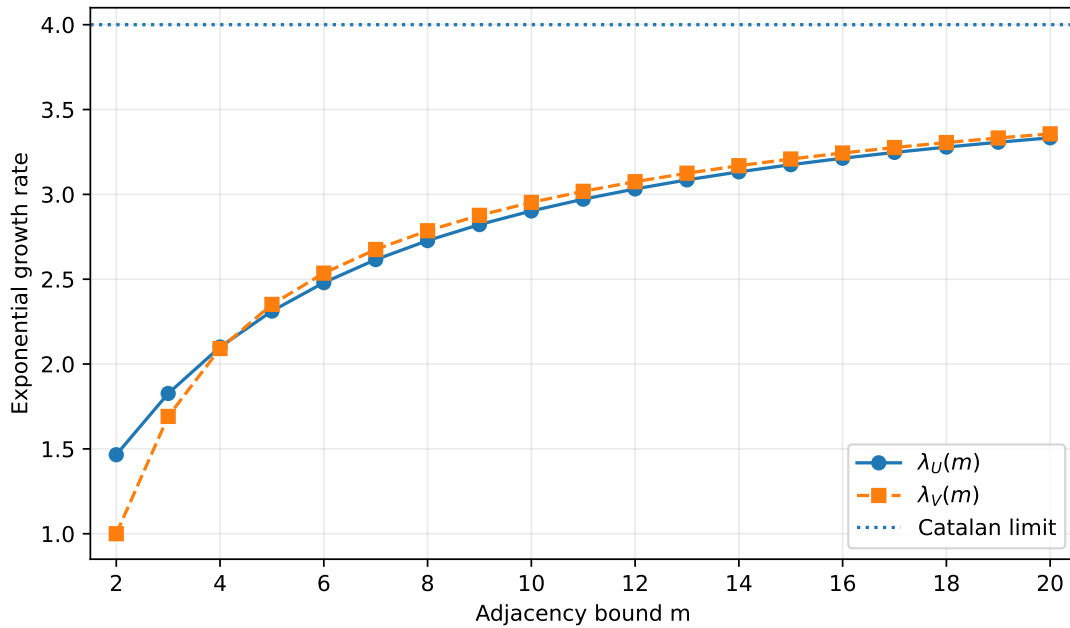


Figure 1: The two component rates for $2 \leq m \leq 20$. Both increase, and the component attaining the maximum changes between $m = 4$ and $m = 5$. The proof of strict growth does not depend on this crossover pattern.

8.3 Large- m scalar comparisons

The scalar bounds can be evaluated without matrices of order m^2 . The implementation stores normalized Catalan coefficients $C_k 4^{-k}$, avoiding enormous intermediate Catalan integers in this

numerical calculation. It solves for $t = m \log(4x)$, not for a root whose distance from $1/4$ is tiny.

Table 3: Approximate evaluation of the rigorous scalar root bounds. The lower column is $\max_{d \in \{1,2,4,8\}} s_{m,d}^{-1}$; the upper is q_m^{-1} . These decimal evaluations are floating-point results, not outward-rounded interval certificates.

m	Lower-bound value	Upper-bound value
10	2.7001330054	3.2847596398
20	3.2358742549	3.5390478302
50	3.6448786614	3.7589497022
100	3.8065234928	3.8581087164
1000	3.9751398963	3.9790979985
10000	3.9969422752	3.9972999790
100000	3.9996392834	3.9996731933
1000000	3.9999586173	3.9999618975

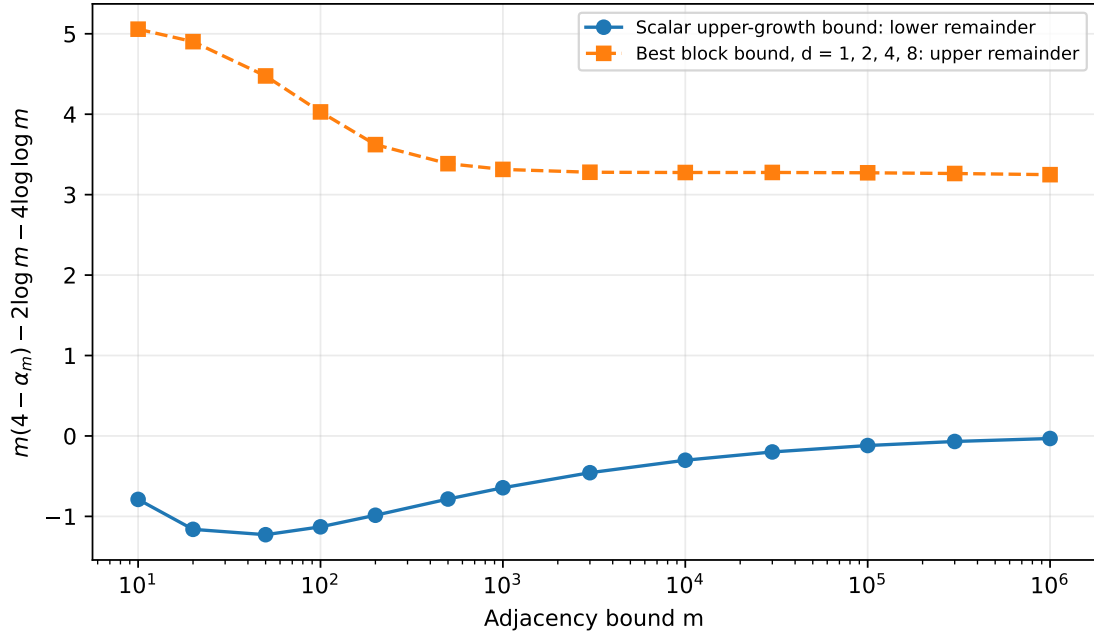


Figure 2: Numerical evaluations of the scalar bounds on the centered remainder E_m . The plotted upper envelope uses only $d = 1, 2, 4, 8$. Neither curve is the true remainder, and the asymptotic constants in (3) are not finite- m interval bounds.

8.4 What was actually checked

The bundled `verification_log.txt` records a successful complete run of the standard-library verifier. Its checks include direct triple-definition testing of all permutations through length seven against the Catalan generator; the first-entry formula for every Catalan avoider through length ten; 2,840 cumulative endpoint counts for $m = 1, \dots, 8$; and 53,534 weighted-edge comparisons under the shift for $m = 2, \dots, 30$. Both component graphs are also tested for strong connectivity in those finite cases.

The verifier checks coefficientwise scalar majorants and block lower bounds through $n = 100$ for $m = 2, \dots, 12$ and the admissible tested values of d . It validates the 38 rational Perron intervals using 3,458 strict integer row inequalities, then compares their endpoints as exact fractions. Finally, it checks all four full polynomial-matrix certificates and compares the resulting

generating-function coefficients with the recurrence through $n = 200$.

For certificate checking, the verifier reconstructs states and weighted edges separately, using Catalan convolutions in place of the binomial formula used by the numerical matrix builder. It also checks agreement between these two implementations. This reduces the risk that a common implementation error merely reproduces itself. Nevertheless, finite computational checks are supplements to, not replacements for, the universal arguments in sections 4, 6 and 7.

9 What is resolved, and what remains

The strict inequality $\alpha_{m+1} > \alpha_m$ is proved for every positive integer m . Together with the existing fixed-bound results, it completes the growth-constant conjecture selected here. The scalar comparison proof also determines the leading constant and next logarithmic term in the approach to four.

Several nearby questions are not resolved by these arguments. The first is whether $\lambda_V(m) > \lambda_U(m)$ for every $m \geq 5$; the reverse inequality holds for $m = 2, 3, 4$. This is the separate component-comparison problem formulated in [2, Problem P3]. Strict increase in both sequences is compatible with crossings, so it cannot settle their ordering. The finite certificates here establish the pattern only through $m = 20$. [Added 28 September 2026, batch 38: this question remains open. Part II shows only that a dominant V can lead by at most $(\lambda_V(m) - \lambda_U(m))_+ = O((\log m)^2/m^2)$ (corollary 15.4), which is why the ordering is invisible to every inverse-logarithmic term; see section 19.]

A second question is whether E_m converges, and, if so, to which constant. The scalar majorant discards endpoint compatibility, while the lower construction permits only a special recoverable family of blocks. Their limiting constants leave a window of width $4 \log 2$. Closing that gap requires controlling more of the endpoint information, not merely taking more terms of the same scalar expansions. A growing- d analysis could improve finite bounds, but our argument deliberately takes d fixed and then sends d to infinity only after a limsup. [Added 28 September 2026, batch 38: the preceding paragraph describes the report at its pinned state 6c9175a54. The question is answered in Part II: $E_m \rightarrow 2 \log \pi$ (corollary 15.3). As anticipated here, the upper side keeps last-endpoint information, through a Catalan-weighted positive vector (lemma 12.2), and the lower side lets the cutoff grow, $d = \lceil \log_2 m \rceil$, using a tail bound uniform in d (lemma 13.2 and theorem 15.2).]

Finally, the minimal recurrence order and uniform denominator factorization of $A_m(x)$ remain outside the scope of this report. The exact $m \leq 4$ artifacts are useful examples, not a claimed solution of that general algebraic problem. The new strict-growth argument is valuable precisely because it avoids needing such a formula.

Part II. The exact constant: an endpoint renewal comparison

Sections 1–9 and the appendices form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition. Part II was added on 28 September 2026 in batch 38 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the question Part I left open in section 9: E_m converges, and its limit is $2\log \pi$. Section 10 records the provenance and fixes the notation of the addition.

10 Provenance, conventions and the answer

10.1 One manuscript, one addition

Part II is batch-38 manuscript 04, *Closing the Catalan growth-constant gap: Endpoint renewal comparison for adjacency-bounded 132-avoiding permutations*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated September 28, 2026 (23 pages as delivered). It arrived as the archive `Closing_Catalan_Growth_Gap.zip` (inner directory `Adjacency_Growth_Constant/`) in ProveIt commit `fbba58593` and was placed in `938b2f74e`. It was written against ProveIt commit `6c9175a54`, at which Part I’s `article.tex` had the Git blob `5ec75ee4`; that blob is unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code and data are shipped beside Part I’s files with the prefix `02-centered-remainder-` (Part I’s unprefix files count as 01); its manuscript, README and PDF are not shipped and survive in the arrival commit. Its two figures were added to `figures/` in the write phase (section 20.5).

The manuscript contributes, in the order printed here: a Catalan-weighted spectral upper bound $\alpha_m < \beta_m$ (section 12); a uniform first-deficiency tail bound and a lower bound for the first component rate (section 13); a uniform endpoint expansion of the truncated Catalan root (section 14); a corridor of width $O((\log m)^2/m^2)$ from a logarithmically growing cutoff, and with it the exact constant (section 15); an all-orders inverse-logarithmic expansion (section 16); exact checks and numerical illustrations (section 17); an $m = 2$ example with a non-implication (section 18); nine research directions (section 19); and its proof boundaries and provenance (section 20). Every result, proof, example, remark, limitation and question of the manuscript is printed. The manuscript re-derives the finite endpoint structure, the first-entry lemma and the injective block construction of Part I; those are printed once, in Part I, and credited where Part II uses them.

Status (manuscript 04). The manuscript presents mathematical proofs of a repository-open asymptotic question and stronger approximation results. It is not externally refereed and contains no new Lean certification. Novelty is assessed against the inspected repository snapshot and the primary papers cited here, not against an exhaustive literature census. Nothing in Part II is formalized in any proof assistant.

10.2 Notation of Part II

Part I’s symbols keep their meanings: $a_n^{(m)}$, A_m , α_m , C_j , C , B_m , $c_{k,p}$, W_m , U_m , V_m , $r_C(m)$, $\lambda_C(m)$, $h(k, j)$, $K_{m,d}$, $s_{m,d}$, $\leq$, and the centered remainder E_m of theorem 1.3. The manuscript uses the same symbols with the same meanings. Its new symbols, and the ones renamed here to avoid a clash with Part I, are listed in table 4.

No normalization changed: every renamed symbol denotes exactly the manuscript’s quantity. Logarithms are natural.

Table 4: Symbols of Part II. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
$K_m(x)$ (one index)	$\sum_{k=1}^m C_{k-1} x^k$	Part I’s block polynomial $K_{m,d}$ (two indices). In particular K_{m-d} is the Catalan polynomial truncated at $m-d$, and $K_{m,d} \leq K_{m-d}$. Part I’s majorant is $R_m = x + K_m$.
$r_m, \beta_m = r_m^{-1}$	positive root of $K_m(r_m) = 1$	the component roots $r_U(m), r_V(m)$, and the local r_m inside lemma 6.1. Since $R_m = x + K_m$, $q_m < r_m$ and $\beta_m < q_m^{-1}$.
t_m	$m \log(4r_m) = m \log(4/\beta_m)$	Part I’s $t_m^* = m \log(4/\alpha_m)$; $t_m < t_m^*$ for $m \geq 2$.
b_k	$C_{k-1} 4^{-k}$ for all $k \geq 1$, so $b_1 = \frac{1}{4}$	the coefficients of R_m in section 7, where $b_1 = \frac{1}{2}$.
Λ (renamed from L)	$\frac{1}{2} \log m$	Part I’s $L_m = \frac{1}{2} \log m + \log \log m$.
z	$\log \Lambda + \log(2\sqrt{\pi}) = \log \log m + \frac{1}{2} \log \pi$	—
$\mathcal{P}_j(z)$ (renamed from P_j)	correction polynomials, proposition 16.1	the numerator polynomials P_3, P_4 of section A.
$\mathcal{B}(w), \mathcal{B}_N(t)$ (renamed from B_N)	the formal series $\sum_j (3/2)_j w^j$ and its truncation $\sum_{j < N} (3/2)_j t^{-j}$	B_m (thresholds), B_d of (23), B of lemma 6.1.
$H_q(x)$	$\sum_{j=0}^q C_j x^j$	the expanded adjacency matrix H in the proof of proposition 3.3.
ε_d	$(d+2)2^{-d}$	Part I’s a_d ; $1 - a_d = \varepsilon_d/2$ (remark 13.3).
$d = d_m$	$\lceil \log_2 m \rceil$ in section 15	Part I, where d is fixed while $m \rightarrow \infty$.
Δ_m (renamed from T_m)	mass of the last d tilted terms	Part I’s endpoint counts $T_{p,q}^{(m)}(n)$.
C_*, c_* (renamed from C, c)	absolute constants	Catalan numbers C_j ; the component letter $C \in \{U, V\}$.
W_{U_m}, W_{V_m} (manuscript: W_U, W_V)	component matrices of Part I	—

10.3 The question and the answer

Part I proved $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$ (theorem 1.2) and, for the centered remainder $E_m = m(4 - \alpha_m) - 2 \log m - 4 \log \log m$, the window (3) of width $4 \log 2$. Its concluding discussion asks whether E_m converges and, if so, to which constant. That is the question resolved here. It was an open question in the inspected report (Part I), not a claim that every open problem in the original enumeration papers has been settled.

The older upper comparison was based on $R_m(x) = x + \sum_{k=1}^m C_{k-1}x^k$. The extra x accounts for appending a maximum when endpoint restrictions are discarded. One cannot simply delete that term from the counting recurrence. Instead, we retain the last-endpoint information and assign a different positive weight to each endpoint threshold. A Catalan convolution then incorporates the append transition without paying for it as an independent unconstrained scalar choice. Part I's diagnosis in section 9, that closing the window needs more endpoint information than the scalar series carry, is exactly what this weighting supplies on the upper side; on the lower side a cutoff d growing with m is used, made legitimate by a bound uniform in d .

Write

$$C_j = \frac{1}{j+1} \binom{2j}{j}, \quad K_m(x) = \sum_{k=1}^m C_{k-1}x^k, \quad K_m(r_m) = 1, \quad \beta_m = r_m^{-1}. \quad (36)$$

The polynomial K_m is strictly increasing on the positive real axis. Thus r_m is well defined. Since $K_m(1/4) < \sum_{k \geq 1} C_{k-1}4^{-k} = 1/2$, we have $r_m > 1/4$. For $m \geq 2$, $K_m(1) > 1$, so $r_m < 1$.

Theorem 10.1 (Finite renewal upper bound). *For every $m \geq 2$,*

$$\alpha_m < \beta_m < 4.$$

For $m = 1$, $\alpha_1 = \beta_1 = 1$.

Theorem 10.2 (Quantitative scalar universality). *As $m \rightarrow \infty$,*

$$0 \leq \beta_m - \alpha_m \leq \beta_m - \lambda_U(m) = O\left(\frac{(\log m)^2}{m^2}\right), \quad (37)$$

where $\lambda_U(m)$ is the first cyclic-component growth rate of (13). In particular,

$$\lim_{m \rightarrow \infty} E_m = 2 \log \pi. \quad (38)$$

For the all-orders statement, put

$$\Lambda = \frac{1}{2} \log m, \quad z = \log \Lambda + \log(2\sqrt{\pi}) = \log \log m + \frac{1}{2} \log \pi. \quad (39)$$

Theorem 10.3 (All inverse-logarithmic orders). *There are explicitly computable polynomials $\mathcal{P}_j \in \mathbb{Q}[z]$, of degree j , such that for every fixed integer $J \geq 0$,*

$$m(4 - \alpha_m) = 4\Lambda + 4z + 4 \sum_{j=1}^J \frac{\mathcal{P}_j(z)}{\Lambda^j} + O_J\left(\frac{(1 + |z|)^{J+1}}{\Lambda^{J+1}}\right). \quad (40)$$

The first three are

$$\begin{aligned} \mathcal{P}_1(z) &= z - \frac{3}{2}, \\ \mathcal{P}_2(z) &= -\frac{1}{2}z^2 + \frac{5}{2}z - \frac{33}{8}, \\ \mathcal{P}_3(z) &= \frac{1}{3}z^3 - 3z^2 + \frac{43}{4}z - 15. \end{aligned} \quad (41)$$

The same expansion holds with α_m replaced by $\lambda_U(m)$ or β_m .

Since $4\Lambda + 4z = 2 \log m + 4 \log \log m + 2 \log \pi$, the case $J = 0$ is again $E_m \rightarrow 2 \log \pi$: the upper endpoint of Part I's window (3). Part II refutes nothing in Part I; it sharpens the window to a point.

10.4 What is inherited and what is added

Table 5: Attribution and scope. “Here” means proved in Part II (manuscript 04).

Ingredient or conclusion	Provenance
Endpoint recurrence and two cyclic components	Mayama–Akita; rederived in Part I and again in manuscript 04
Strict increase of α_m and the window (3)	Part I; not claimed anew
Injective skew blocks with fixed endpoint cutoff	Part I; recalled
Catalan-weighted spectral upper bound $\alpha_m < \beta_m$	Here
Uniform tail control and $O((\log m)^2/m^2)$ approximation	Here
Exact constant $2 \log \pi$ and the transferred all-orders expansion	Here
Ordering of U and V for every $m \geq 5$	Not resolved here

The inspected repository path, commit and source blob are recorded in section 20.4. This matters because the repository is an active research collection: an unpinned statement about what it contains would not be a reproducible novelty comparison.

11 The finite interface

This section collects the input to the new comparison. Its finite-state structure is established work [2] and is proved in Part I; manuscript 04 rederives it, to make its argument independent of undocumented implementation details, and its proofs coincide with Part I’s except where a second route is marked below.

The endpoint recurrence is proposition 2.1. In the language of edges, $W_m(x)$ of (9) has a split edge

$$(p, q) \longrightarrow (m - k, q - k) \quad \text{with weight } c_{k,p} x^k, \quad (42)$$

and an append edge

$$(p, q) \longrightarrow (p - 1, m - 1) \quad \text{with weight } x, \quad (43)$$

invalid negative thresholds being omitted. By (6),

$$0 \leq c_{k,p} \leq C_{k-1}, \quad c_{k,p} \leq c_{k,p+1}, \quad c_{k,\infty} = C_{k-1}. \quad (44)$$

Proposition 11.1 (Spectral interface). *For $m \geq 2$, both $W_{U_m}(x)$ and $W_{V_m}(x)$ are irreducible for $x > 0$. Their spectral radii are continuous and strictly increasing from zero to infinity. With $r_C(m)$ and $\lambda_C(m)$ as in (13),*

$$\alpha_m = \max\{\lambda_U(m), \lambda_V(m)\}, \quad \lambda_U(m) > 1. \quad (45)$$

Proof. This is lemma 3.1, the paragraph after lemma 3.2, and proposition 3.3. Manuscript 04’s Appendix A gives the same graph and transfer-matrix proof: the same strong-connectivity induction on the gap $p - q$, the same classification of cyclic components, the same expansion of every edge into unit-length edges, and the same closed walks of lengths two and five for primitivity. Its argument for $\lambda_U(m) > 1$ differs and is printed next. $\square$

Second proof of $\lambda_U(m) > 1$ (manuscript 04). Let $e_U = (m - 1, \infty)$. At $x = 1$, the column of W_{U_m} indexed by e_U consists of ones, from the $k = 1$ edges. Consequently $W_{U_m}(1)\delta_{e_U} = \mathbf{1} \geq \delta_{e_U}$, strictly in another coordinate when $m \geq 2$. A positive left Perron vector gives $\text{spr}(W_{U_m}(1)) > 1$, and hence $r_U(m) < 1$. $\square$

Remark 11.2 (Constants depend on m). All constants in the proof of [proposition 3.3](#) (the c , C and D of its last paragraph) may depend on m ; no joint-limit uniformity is claimed, in Part I or here. The interface is all that Part II uses: not explicit determinants, a conjectural ordering of components, or a simple-pole assertion about A_m .

We will repeatedly use two elementary consequences of Perron–Frobenius positivity.

- Lemma 11.3** (Positivity tools). (i) *If M is irreducible and nonnegative, and $v > 0$ satisfies $Mv \leq v$ with a strict inequality in at least one coordinate, then $\text{spr}(M) < 1$.*
 (ii) *For any nonnegative matrix, the spectral radius is at least its minimum row sum, and it is monotone under entrywise increase and under principal extension.*

Proof. For (i), multiply by a positive left Perron vector u : $\text{spr}(M)u^\top v = u^\top Mv < u^\top v$. For (ii), the row-sum bound follows by iterating $M\mathbf{1} \geq a\mathbf{1}$; the monotonicity follows by comparing powers. The mechanism of (i) is the one used in [lemma 3.2](#). $\square$

12 The endpoint renewal comparison

The essential new finite observation is a positive vector adapted to the last threshold. Before applying it, it is useful to isolate its algebraic mechanism.

12.1 A renewal system with a reset

Lemma 12.1 (Renewal–reset identity). *Let $w_1, \dots, w_{m-1} \geq 0$ and $a \geq 0$. On indices $0, \dots, m-1$, let L be the strictly lower triangular matrix with $L_{q,q-k} = w_k$ for $1 \leq k \leq q$, and put*

$$N = L + a\mathbf{1}e_{m-1}^\top.$$

Define $u_0 = 1$, $u_n = \sum_{k=1}^n w_k u_{n-k}$ for $1 \leq n < m$, and $H_q = \sum_{j=0}^q u_j$. Then

$$(I - L)H = \mathbf{1}, \tag{46}$$

$$NH = H + (aH_{m-1} - 1)\mathbf{1}, \tag{47}$$

$$\det(I - N) = 1 - aH_{m-1}. \tag{48}$$

Proof. For each q ,

$$\sum_{k=1}^q w_k H_{q-k} = \sum_{n=1}^q \sum_{k=1}^n w_k u_{n-k} = \sum_{n=1}^q u_n = H_q - 1.$$

This proves (46); the reset term gives (47). Since L is strictly triangular, $\det(I - L) = 1$. The determinant identity for a rank-one update, or direct elimination of $(I - L)$, gives

$$\det(I - N) = \det(I - L)(1 - ae_{m-1}^\top(I - L)^{-1}\mathbf{1}) = 1 - aH_{m-1}.$$

All identities also hold over a commutative coefficient ring when the nonnegativity assertions are omitted. $\square$

For the Catalan application, take $w_k = C_{k-1}x^k$ and $a = x$. The Catalan recurrence gives $u_j = C_j x^j$. Thus

$$H_q(x) = \sum_{j=0}^q C_j x^j, \quad xH_{m-1}(x) = K_m(x). \tag{49}$$

Equivalently, the formal identity $C(x) = 1 + xC(x)^2$ gives $(1 - xC(x))^{-1} = C(x)$. The finite version we need is

$$\sum_{k=1}^q C_{k-1}x^k H_{q-k}(x) = H_q(x) - 1. \tag{50}$$

There is no convergence issue in this polynomial identity.

12.2 A positive vector for the true two-endpoint component

Lemma 12.2 (Last-threshold supersolution). *For a state $(p, q) \in V_m$, set $v_{p,q}(x) = H_q(x)$. For every $x > 0$,*

$$(W_{V_m}(x)v(x))_{p,q} \leq H_q(x) - 1 + K_m(x). \quad (51)$$

At $x = r_m$, at least one coordinate inequality is strict.

Proof. A valid split from (p, q) has $k \leq q$, and its target has last threshold $q - k$. We may bound $c_{k,p}$ by C_{k-1} and include every $1 \leq k \leq q$. If an append edge exists, its target has last threshold $m - 1$; including such an edge even when it is absent can only increase the result. Therefore

$$\begin{aligned} (W_{V_m}v)_{p,q} &\leq \sum_{k=1}^q C_{k-1}x^k H_{q-k}(x) + xH_{m-1}(x) \\ &= H_q(x) - 1 + K_m(x), \end{aligned}$$

by (50). At the state $(0, m - 1)$, which belongs to V_m for $m \geq 2$, there is no append edge. The positive term $xH_{m-1}(x)$ was consequently added strictly. This proves strictness, in fact for every $x > 0$. $\square$

Proof of theorem 10.1. At $x = r_m$, lemma 12.2 gives $W_{V_m}v \leq v$, with a strict coordinate and $v > 0$. Irreducibility and lemma 11.3(i) imply $\text{spr}(W_{V_m}(r_m)) < 1$.

In U_m , append edges leave the component. Its row at first threshold p has sum $\sum_{k=1}^m c_{k,p}r_m^k \leq K_m(r_m) = 1$. The row $p = 0$ has sum $r_m < 1$, since only the $k = 1$ coefficient survives. Thus $\text{spr}(W_{U_m}(r_m)) < 1$ by the same positivity argument, now with vector $\mathbf{1}$.

Strict increase of each spectral radius gives $r_U(m), r_V(m) > r_m$. Taking reciprocals and using (45) proves $\alpha_m < \beta_m$. The already established $r_m > 1/4$ gives $\beta_m < 4$. For $m = 1$, the only permutations of length at least two with jumps of absolute value one are the increasing and decreasing permutations; a change of direction would repeat a value. Hence $\alpha_1 = 1 = \beta_1$. $\square$

Remark 12.3 (Comparison with Part I's majorant). Since $R_m = x + K_m$, one has $R_m(r_m) = 1 + r_m > 1$, so $q_m < r_m$ and $\beta_m < q_m^{-1}$ for every $m \geq 1$: theorem 10.1 is strictly sharper than proposition 5.1. At $m = 100$, for instance, the upper bound drops from $3.8581 \dots$ (table 3) to $3.8201 \dots$ (table 7). Part I's bound is coefficientwise, theorem 10.1 only spectral; see the next remark.

Remark 12.4 (What this comparison does not say). The scalar polynomial is not asserted to be a denominator of A_m , and $\det(I - W_{V_m})$ is not asserted to equal $1 - K_m$. The determinant identity (48) belongs to the *majorizing reset system*, not to the full endpoint system. Nor does the proof assert the coefficientwise inequality $A_m(x) \leq (1 - K_m(x))^{-1}$; it is false (section 18). Our comparison is a spectral one, witnessed by a positive vector. These distinctions prevent a valid growth bound from being misread as a solution of state minimization or exact enumeration.

13 A uniform lower approximation from recoverable blocks

The block construction itself is that of proposition 5.2 in Part I. The additional ingredients of manuscript 04 are a component version of the lower bound, an estimate uniform in the cutoff, and its use with a cutoff growing with m .

13.1 Block and component lower bounds

Fix integers $1 \leq d < m$, and let $K_{m,d}$ and $s_{m,d}$ be as in (18) and proposition 5.2:

$$K_{m,d}(x) = \sum_{k=1}^{m-d} c_{k,d}x^k, \quad K_{m,d}(s_{m,d}) = 1. \quad (52)$$

The coefficient of x is one, so a unique positive root exists.

Proposition 13.1 (Block and component lower bounds). *For every $1 \leq d < m$,*

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x), \quad s_{m,d}^{-1} \leq \lambda_U(m) \leq \alpha_m < \beta_m. \quad (53)$$

Proof. The coefficientwise inequality is proposition 5.2: skew sums of blocks ending at their maximum, with the boundary jump $k + j \leq m$, and injectivity because the skew cuts recover the block boundaries. Manuscript 04 reproves it by the same argument.

For the sharper component statement, restrict $W_{U_m}(x)$ to first thresholds $p \in \{d, \dots, m-1\}$. For every $1 \leq k \leq m-d$, the split target $m-k$ is in that subset and its coefficient satisfies $c_{k,p} \geq c_{k,d}$. Every row sum of this principal submatrix is at least $K_{m,d}(x)$. At $x = s_{m,d}$ its spectral radius is at least one. Principal comparison (lemma 11.3(ii)) implies $\text{spr}(W_{U_m}(s_{m,d})) \geq 1$, so $r_U(m) \leq s_{m,d}$ and $\lambda_U(m) \geq s_{m,d}^{-1}$. The remaining inequalities are (45) and theorem 10.1. $\square$

13.2 First-entry counts and a uniform tail

The first-entry formula is lemma 5.3: $h(k, j) = \frac{j}{k-1} \binom{2k-j-3}{k-2}$ for $1 \leq j < k$, with generating function (21). Manuscript 04 reproves it with the same slot decomposition and the same Lagrange inversion. Consequently $c_{k,d} = \sum_{j=1}^{\min(d, k-1)} h(k, j)$ for $k \geq 2$, as in (7).

Lemma 13.2 (Uniform deficiency-tail estimate). *For all $d \geq 1$ and $k \geq 1$,*

$$0 \leq C_{k-1} - c_{k,d} \leq \varepsilon_d C_{k-1}, \quad \varepsilon_d = (d+2)2^{-d}. \quad (54)$$

Hence, whenever $1 \leq d < m$ and $\varepsilon_d < 1$,

$$(1 - \varepsilon_d)K_{m-d}(x) \leq K_{m,d}(x) \leq K_{m-d}(x) \quad (x \geq 0). \quad (55)$$

The corresponding derivative inequalities also hold for $x \geq 0$.

Proof. There is nothing to prove if $k = 1$ or $k \leq d+1$. For the remaining cases, formula (21) gives

$$\frac{h(k, j)}{C_{k-2}} = j \prod_{r=0}^{j-2} \frac{k-2-r}{2k-4-r} \leq \frac{j}{2^{j-1}}. \quad (56)$$

The product is empty for $j = 1$. Each displayed factor is at most $1/2$. For a missing deficiency $j \geq d+1 \geq 2$, we have $k \geq 3$ and

$$\frac{C_{k-2}}{C_{k-1}} = \frac{k}{4k-6} \leq \frac{1}{2}.$$

Thus $h(k, j)/C_{k-1} \leq j2^{-j}$. Summing missing deficiencies and extending the tail to infinity gives

$$\frac{C_{k-1} - c_{k,d}}{C_{k-1}} \leq \sum_{j=d+1}^{\infty} \frac{j}{2^j} = \frac{d+2}{2^d}.$$

For the final equality, differentiate the finite-tail geometric sum or subtract its successive terms. Multiplying coefficientwise by x^k and summing proves (55). Differentiation preserves a coefficientwise inequality between these nonnegative polynomials. $\square$

The identity in (56) is Part I's (24); the inequality is new.

Remark 13.3 (Merge note: the uniform bound against Part I's limit). By (25) and $C_{k-1}4^{-k} \sim (4\sqrt{\pi})^{-1}k^{-3/2}$, the missing fraction $1 - c_{k,d}/C_{k-1}$ tends to $1 - a_d = (d+2)2^{-(d+1)} = \varepsilon_d/2$ as $k \rightarrow \infty$. The uniform bound ε_d of lemma 13.2 therefore loses at most a factor two against the limit, the factor coming from $C_{k-2}/C_{k-1} \leq \frac{1}{2}$ where the limit is $\frac{1}{4}$. This observation is the merge's, not the manuscript's.

The role of uniformity is decisive. A fixed- d asymptotic for $c_{k,d}$, such as (25), would not justify substituting $d = d(m)$. Inequality (54) holds simultaneously for all k, d , so the growing cutoff used below requires no interchange of unproved uniform limits.

14 The truncated Catalan root at all orders

The remaining analysis begins with the scalar root alone. Normalize its coefficients by

$$b_k = C_{k-1}4^{-k}, \quad \sum_{k \geq 1} b_k = \frac{1}{2}, \quad b_k = \frac{1}{4\sqrt{\pi}} k^{-3/2} (1 + O(k^{-1})). \quad (57)$$

The last estimate follows from Stirling's formula. In particular, $b_k \leq C_* k^{-3/2}$ with one absolute constant C_* . If

$$t_m = m \log(4r_m), \quad (58)$$

then

$$\sum_{k=1}^m b_k e^{t_m k/m} = 1, \quad \beta_m = 4e^{-t_m/m}. \quad (59)$$

The tail is summable but has total mass only $1/2$. A positive tilt must supply the missing half. It does so in a thin layer near $k = m$.

14.1 A uniform endpoint expansion

For $a \in \mathbb{R}$ and $j \geq 0$, write $(a)_0 = 1$ and $(a)_j = a(a+1) \cdots (a+j-1)$.

Lemma 14.1 (Tilted Catalan endpoint expansion). *Fix $M > 0$ and an integer $N \geq 1$. Uniformly for*

$$t = \frac{1}{2} \log m + \log \log m + s, \quad |s| \leq M, \quad (60)$$

as $m \rightarrow \infty$,

$$\sum_{k=1}^m b_k e^{tk/m} = \frac{1}{2} + \frac{e^t}{4\sqrt{\pi m} t} \left(\sum_{j=0}^{N-1} \frac{(3/2)_j}{t^j} + O_{M,N}(t^{-N}) \right). \quad (61)$$

Proof. Split the sum at $m/2$. For the lower half, the unweighted partial sum is $1/2 + O(m^{-1/2})$. Since $e^u - 1 \leq ue^u$ for $u \geq 0$, its tilt increment is at most

$$\begin{aligned} \sum_{k \leq m/2} b_k (e^{tk/m} - 1) &\leq \frac{C_* t}{m} e^{t/2} \sum_{k \leq m/2} k^{-1/2} \\ &= O\left(\frac{te^{t/2}}{\sqrt{m}}\right) = O_M\left(m^{-1/4}(\log m)^{3/2}\right). \end{aligned}$$

This is smaller than every fixed negative power of t . Thus the lower half contributes $1/2$ plus an error that can be absorbed into the remainder in (61).

On the upper half, Stirling's estimate is uniform:

$$b_k = \frac{1}{4\sqrt{\pi}} k^{-3/2} (1 + O(m^{-1})).$$

A Riemann-sum comparison, using that $y^{-3/2}e^{ty}$ is increasing on $[1/2, 1]$ for all sufficiently large t , gives

$$\begin{aligned} \sum_{m/2 < k \leq m} b_k e^{tk/m} &= \frac{1}{4\sqrt{\pi m}} \int_{1/2}^1 y^{-3/2} e^{ty} dy + O\left(\frac{e^t}{m^{3/2}}\right) \\ &= \frac{e^t}{4\sqrt{\pi m} t} \int_0^{t/2} (1 - v/t)^{-3/2} e^{-v} dv + O\left(\frac{e^t}{m^{3/2}}\right). \end{aligned}$$

A possible half-integer lower endpoint changes only the same error term. For $0 \leq v \leq t/2$, Taylor's theorem with a uniform remainder gives

$$(1 - v/t)^{-3/2} = \sum_{j=0}^{N-1} \frac{(3/2)_j}{j!} (v/t)^j + O_N((v/t)^N).$$

Integrating each monomial against e^{-v} and extending its upper limit to infinity produces $j!$; the extension error is exponentially small in t . The integral is therefore $\sum_{j=0}^{N-1} (3/2)_j t^{-j} + O_N(t^{-N})$.

In the window (60), the prefactor $e^t/(\sqrt{m} t)$ is bounded above and below by positive constants depending on M . The relative Riemann error is $O(t/m)$, and all lower-half errors are smaller than any fixed t^{-N} . Combining the estimates proves the uniform expansion. $\square$

Corollary 14.2 (Scalar root location). *As $m \rightarrow \infty$,*

$$t_m = \frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi + o(1). \quad (62)$$

In particular, $t_m = O(\log m)$, $e^{t_m} = O(\sqrt{m} \log m)$, and $r_m \rightarrow 1/4$.

Proof. For a fixed real s , the $N = 1$ case of lemma 14.1 gives

$$\sum_{k=1}^m b_k \exp\left(\frac{k}{m} \left[\frac{1}{2} \log m + \log \log m + s\right]\right) \rightarrow \frac{1}{2} + \frac{e^s}{2\sqrt{\pi}}.$$

The limit equals one precisely at $s_0 = \frac{1}{2} \log \pi$. For every $\delta > 0$, its values at $s_0 - \delta$ and $s_0 + \delta$ lie strictly on opposite sides of one. The finite sum is strictly increasing in t , so its root lies between those trial values for all sufficiently large m . Let $\delta \downarrow 0$. The remaining assertions follow directly from (62) and (58). $\square$

Second proof (merge note, from Part I). Apply lemma 6.1 to $b_k = C_{k-1} 4^{-k}$, $h = 0$: then $B = \frac{1}{2}$, $c = 1/(4\sqrt{\pi})$, and $\log \frac{1-B}{2c} = \log \sqrt{\pi}$, which is (62). Part I applied the same lemma to the coefficients of R_m ($B = \frac{3}{4}$), which gave the lower endpoint of its window; the extra x of R_m alone accounts for the loss of $4 \log 2$ there, as $4 \log \frac{1-3/4}{1-1/2} = -4 \log 2$. $\square$

Remark 14.3 (An immediate resolution using the earlier window). Already theorem 10.1 and corollary 14.2 imply $\liminf E_m \geq 2 \log \pi$. Combined with the upper bound on $\limsup E_m$ in (3), this closes Part I's question. The next section proves a substantially stronger quantitative result and provides an independent lower argument rather than relying on that asymptotic window.

15 A logarithmically growing cutoff gives a sharp corridor

Take, for all sufficiently large m ,

$$d = d_m = \lceil \log_2 m \rceil, \quad \varepsilon_d = (d+2)2^{-d} = O(\log m/m). \quad (63)$$

In particular, $d < m$, $d = O(\log m)$, and $\varepsilon_d < 1$.

Lemma 15.1 (Value deficit and slope). *At the scalar root r_m , with d as in (63),*

$$0 \leq 1 - K_{m,d}(r_m) = O\left(\frac{(\log m)^2}{m}\right), \quad (64)$$

$$K'_{m,d}(r_m) \geq c_* m \quad \text{for all sufficiently large } m, \quad (65)$$

for an absolute constant $c_ > 0$.*

Proof. Set $t = t_m$. The lost last d terms of $K_m(r_m)$ satisfy

$$\begin{aligned}\Delta_m &:= K_m(r_m) - K_{m-d}(r_m) = \sum_{k=m-d+1}^m b_k e^{t_k/m} \\ &\leq C_* d(m-d)^{-3/2} e^t = O\left(\frac{d \log m}{m}\right) = O\left(\frac{(\log m)^2}{m}\right),\end{aligned}\tag{66}$$

using corollary 14.2. Since $K_m(r_m) = 1$, lemma 13.2 gives

$$K_{m,d}(r_m) \geq (1 - \varepsilon_d)(1 - \Delta_m), \quad 1 - K_{m,d}(r_m) \leq \varepsilon_d + \Delta_m.$$

This proves (64).

The lower-half estimate in the proof of lemma 14.1, now evaluated at t_m , shows

$$\sum_{k \leq m/2} b_k e^{t_m k/m} \rightarrow \frac{1}{2}.$$

The whole sum is one, and $\Delta_m \rightarrow 0$. Consequently

$$\sum_{m/2 < k \leq m-d} b_k e^{t_m k/m} \rightarrow \frac{1}{2}.\tag{67}$$

Every term in this range has $k > m/2$, so

$$K'_{m-d}(r_m) = \frac{1}{r_m} \sum_{k=1}^{m-d} k b_k e^{t_m k/m} \geq \frac{m}{2r_m} \sum_{m/2 < k \leq m-d} b_k e^{t_m k/m}.$$

As $r_m \rightarrow 1/4$, the right side is bounded below by a positive constant times m . The derivative version of (55), together with $\varepsilon_d \rightarrow 0$, proves (65). $\square$

Theorem 15.2 (Quantitative scalar corridor). *With $d = \lceil \log_2 m \rceil$,*

$$0 \leq s_{m,d} - r_m = O\left(\frac{(\log m)^2}{m^2}\right), \quad 0 \leq \beta_m - s_{m,d}^{-1} = O\left(\frac{(\log m)^2}{m^2}\right).\tag{68}$$

In particular, (37) holds.

Proof. Since $K_{m,d} \leq K_m$, we have $s_{m,d} \geq r_m$. Its derivative is nondecreasing on the positive axis, since all coefficients are nonnegative. Writing $s = s_{m,d}$, integration or the mean-value inequality gives

$$1 - K_{m,d}(r_m) = K_{m,d}(s) - K_{m,d}(r_m) \geq K'_{m,d}(r_m)(s - r_m).$$

Divide the estimate (64) by (65) to obtain the first part of (68). Because $s, r_m \geq 1/4$,

$$0 \leq r_m^{-1} - s^{-1} = \frac{s - r_m}{r_m s} \leq 16(s - r_m),$$

which proves the second part. Finally, use the entire chain (53):

$$0 \leq \beta_m - \alpha_m \leq \beta_m - \lambda_U(m) \leq \beta_m - s_{m,d}^{-1}.\tag{69}$$

Corollary 15.3 (The exact constant). *The centered remainder of theorem 1.3 satisfies $E_m \rightarrow 2 \log \pi$. Equivalently,*

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + 2 \log \pi + o(1)}{m}.\tag{69}$$

Proof. By theorem 15.2 and $\beta_m = 4e^{-t_m/m}$,

$$\begin{aligned} m(4 - \alpha_m) &= 4m(1 - e^{-t_m/m}) + O((\log m)^2/m) \\ &= 4t_m + O((\log m)^2/m). \end{aligned} \quad (70)$$

Here Taylor's theorem applies since $t_m = O(\log m)$. Substitute (62). This proves (69) and theorem 10.2 in full. $\square$

Corollary 15.4 (A one-sided constraint on component competition). *Regardless of which component dominates,*

$$0 \leq \alpha_m - \lambda_U(m) = (\lambda_V(m) - \lambda_U(m))_+ = O\left(\frac{(\log m)^2}{m^2}\right). \quad (71)$$

This is deliberately a one-sided conclusion. We have not proved a corresponding lower approximation for $\lambda_V(m)$ when it is below $\lambda_U(m)$, and do not claim an absolute bound on $|\lambda_V(m) - \lambda_U(m)|$ in that case. The result does show that whenever V wins, its advantage is invisible to every inverse-logarithmic term in the scaled deficit.

16 Inverting the scalar expansion to every logarithmic order

The quantitative corridor is what makes a scalar asymptotic expansion a statement about the actual permutation class. We now derive the coefficients, including their error interpretation.

16.1 Finite asymptotic equations, not a convergent series

Introduce the *formal* power series

$$\mathcal{B}(w) = \sum_{j \geq 0} (3/2)_j w^j. \quad (72)$$

Its coefficients grow factorially. The series has radius of convergence zero and is never evaluated as an ordinary convergent series at $w > 0$. For each fixed $N \geq 1$, instead use the finite polynomial in $1/t$

$$\mathcal{B}_N(t) = \sum_{j=0}^{N-1} (3/2)_j t^{-j}.$$

Substituting $t = t_m$ into lemma 14.1 gives

$$t_m - \log t_m + \log \mathcal{B}_N(t_m) = \frac{1}{2} \log m + \log(2\sqrt{\pi}) + O_N(t_m^{-N}). \quad (73)$$

Indeed the missing mass is $1/2$, so $e^{t_m} \mathcal{B}_N(t_m)/t_m = 2\sqrt{\pi m} (1 + O_N(t_m^{-N}))$. The polynomial $\mathcal{B}_N(t)$ is positive for $t > 0$, and its logarithm is unambiguous.

Formal logarithms are used below only as a finite coefficient-extraction device. To compute a coefficient of degree j , only coefficients through degree j of (72) are needed. Thus the divergent notation organizes an unlimited sequence of finite, justified asymptotic statements; it is not an analytic summation prescription.

16.2 A triangular recursion for all coefficients

Proposition 16.1 (Coefficient recursion). *Set $\mathcal{P}_0$ aside and define $\mathcal{P}_j$ recursively as follows. At step $j \geq 1$, put*

$$T_{j-1}(u, z) = 1 + zu + \sum_{h=1}^{j-1} \mathcal{P}_h(z) u^{h+1},$$

and define

$$\mathcal{P}_j(z) = [u^j] \left\{ \log T_{j-1}(u, z) - \log \mathcal{B} \left(\frac{u}{T_{j-1}(u, z)} \right) \right\}. \quad (74)$$

These are well-defined rational polynomials of degree j , with leading coefficient $(-1)^{j+1}/j$. For every fixed $J \geq 0$,

$$t_m = \Lambda + z + \sum_{j=1}^J \frac{\mathcal{P}_j(z)}{\Lambda^j} + O_J \left(\frac{(1+|z|)^{J+1}}{\Lambda^{J+1}} \right), \quad (75)$$

with Λ, z as in (39).

Proof. Write $u = 1/\Lambda$ and seek $t = \Lambda T$, with

$$T = 1 + zu + \sum_{j=1}^J \mathcal{P}_j(z) u^{j+1}.$$

Since $z = \log \Lambda + \log(2\sqrt{\pi})$, substitution into the formal version of (73) leaves the residual

$$\sum_{j=1}^J \mathcal{P}_j(z) u^j - \log T + \log \mathcal{B}(u/T). \quad (76)$$

The coefficient of u^j in the last two terms depends only on $\mathcal{P}_1, \dots, \mathcal{P}_{j-1}$: $\mathcal{P}_j$ first enters T at order u^{j+1} . Cancelling the coefficient of u^j therefore gives exactly (74), proving existence and uniqueness of the recursion.

Induction shows $\deg \mathcal{P}_j \leq j$. The only term of degree j in z at order u^j comes from $\log(1 + zu)$ and equals $(-1)^{j+1} z^j / j$. Terms involving an earlier $\mathcal{P}_h u^{h+1}$ spend at least one more power of u than their possible degree in z ; terms from $\log \mathcal{B}(u/T)$ have the same strict deficit. Thus the degree is exactly j and its leading coefficient has the stated value.

To justify the error, fix J and take $N \geq J + 2$ in the finite equation (73). For the trial value t_* obtained by truncating (75), we have $T = 1 + O_J((1+|z|)/\Lambda)$ and $z = O(\log \Lambda)$. Finite Taylor expansions of the logarithms, with remainders uniform for $|T - 1| \leq 1/2$, turn the formal cancellations into the numerical residual estimate

$$t_* - \log t_* + \log \mathcal{B}_N(t_*) - \Lambda - \log(2\sqrt{\pi}) = O_J((1+|z|)^{J+1} \Lambda^{-J-1}).$$

Both t_* and t_m lie in a window with bounded shift of the form (60), by corollary 14.2 and $\sum_{j \leq J} \mathcal{P}_j(z) \Lambda^{-j} = o(1)$. The derivative of $t - \log t + \log \mathcal{B}_N(t)$ is

$$1 - \frac{1}{t} + O_N(t^{-2}),$$

so it is at least $1/2$ throughout the interval between them for large m . Apply the mean-value theorem to the residuals of t_* and t_m . The $O_N(t_m^{-N})$ term is smaller than the claimed bound. This proves (75). $\square$

The first four polynomials are

$$\begin{aligned} \mathcal{P}_1(z) &= z - \frac{3}{2}, \\ \mathcal{P}_2(z) &= -\frac{1}{2}z^2 + \frac{5}{2}z - \frac{33}{8}, \\ \mathcal{P}_3(z) &= \frac{1}{3}z^3 - 3z^2 + \frac{43}{4}z - 15, \\ \mathcal{P}_4(z) &= -\frac{1}{4}z^4 + \frac{10}{3}z^3 - \frac{153}{8}z^2 + \frac{223}{4}z - \frac{4461}{64}. \end{aligned} \quad (77)$$

For an independent algebraic check, the formal logarithm begins

$$\log \mathcal{B}(w) = \frac{3}{2}w + \frac{21}{8}w^2 + \frac{69}{8}w^3 + \frac{2529}{64}w^4 + O(w^5). \quad (78)$$

The bundled symbolic verifier constructs the polynomials from the recursion and confirms that (76) vanishes through degree four using exact rational polynomial arithmetic.

16.3 Transfer to the true growth constant

Proof of theorem 10.3. Equation (70) gives $m(4 - \alpha_m) = 4t_m + O((\log m)^2/m)$. For each fixed J , this error is smaller than every fixed power of $1/\log m$, and hence is absorbed by the remainder in (75). Multiply that expansion by four. The same proof applies to $\lambda_U(m)$ by (37). For β_m , the direct Taylor expansion of $4e^{-t_m/m}$ supplies the transfer. $\square$

With $z = \log \log m + \frac{1}{2} \log \pi$, the first three corrections can also be written in powers of $1/\log m$ (the manuscript abbreviates $\log m$ as H , which Part II avoids):

$$\begin{aligned} m(4 - \alpha_m) = & 2 \log m + 4 \log \log m + 2 \log \pi + \frac{8z - 12}{\log m} \\ & + \frac{-8z^2 + 40z - 66}{(\log m)^2} \\ & + \frac{\frac{32}{3}z^3 - 96z^2 + 344z - 480}{(\log m)^3} + O\left(\frac{(1 + |\log \log m|)^4}{(\log m)^4}\right). \end{aligned} \quad (79)$$

The assertion is asymptotic for each fixed truncation order. It does not promise that adding another term improves a numerical approximation at a particular small or moderate value of m .

16.4 Two different error scales

There are two conceptually distinct approximations. The scalar surrogate β_m differs from α_m by at most order $(\log m)^2/m^2$. A fixed truncation of the logarithmic expansion generally has a much larger error, of order a power of $1/\log m$ after multiplying the growth deficit by m . Thus a scalar root can be numerically extremely accurate while its first few asymptotic terms are still inaccurate.

This also explains the component issue. The proof does not need to decide whether $\alpha_m = \lambda_U(m)$ or $\lambda_V(m)$. When V is dominant, corollary 15.4 confines its extra growth to a scale smaller than all the terms in (79). A calculation retaining only those terms has discarded the very information needed to settle that component ordering.

17 Exact checks and numerical evidence

The universal results above are proved analytically. The companion computations have three separate roles: exact checks of finite algebra and combinatorics, exact root enclosures in finite examples, and numerical illustrations. They are deliberately not merged into a single ambiguous notion of verification. File names below are the shipped ones; the scripts themselves still read and write their delivered names, so they are rerun on a copy (section 20.2).

17.1 Exact finite checks

The exact verifier, the standard-library program `code/02-centered-remainder-verify.py`, uses integers and fractions. `Fraction`; it needs no floating-point library. Its successful run is recorded in `data/02-centered-remainder-verification_report.json`. The row check verifies coefficientwise that $H_q - (W_{V_m} v)_{p,q} - (1 - K_m)$ has nonnegative coefficients. This is stronger than testing the

Table 6: Executed exact checks (manuscript 04). Finite ranges do not replace the all- m proofs.

Check	Executed range or count
Direct triple-definition avoidance versus recursive Catalan generation	All permutations for $0 \leq n \leq 7$
First-entry formula against explicit maximum-ending blocks	55 counts, $1 \leq k \leq 10$
Actual V -row polynomial domination by the Catalan witness	4,930 rows, $2 \leq m \leq 30$
Uniform deficiency-tail inequality	9,000 pairs, $k \leq 300$, $d \leq 30$
Restricted U -row coefficient comparisons	67,425 comparisons, $m \leq 30$
Exact scalar-root corridors	Five (m, d) pairs

inequality at a list of decimal values of x . The tail check clears the denominator 2^d and verifies an integer inequality. The direct avoidance test provides a separate combinatorial check of the recursive generator before its blocks are used to test the first-entry formula. The first two rows repeat, with independent code and the same ranges, checks of Part I's verifier (section 8); the other four are new.

The symbolic program `code/02-centered-remainder-verify_expansion.py` uses SymPy but no floating-point coefficients. It verifies (77), (78), and the vanishing residual through degree four. Its output is `data/02-centered-remainder-expansion_verification.json`. Symbolic cancellation proves these finite coefficient identities, not the analytic error estimate; that estimate is the content of lemma 14.1 and proposition 16.1.

17.2 Certified finite corridors

For a positive polynomial P with zero constant term, exact bisection encloses the root of $P(x) = 1$ by fractions ℓ, u satisfying $P(\ell) \leq 1 \leq P(u)$. Apply this separately to K_m and $K_{m,d}$. If the enclosures are $[\ell_r, u_r]$ and $[\ell_s, u_s]$, then

$$\frac{1}{u_s} \leq \alpha_m \leq \frac{1}{\ell_r}. \quad (80)$$

The rational certificate file `data/02-centered-remainder-rational_certificates.json` retains the full fractions and can be regenerated by the exact verifier. The following decimal endpoints are rounded *outwards* from those fractions, not nearest-rounded approximations to unknown values of α_m . These intervals are not advertised as optimal small- m estimates. Their purpose is

Table 7: Rigorous finite intervals from exact scalar bisection.

m	d	Lower endpoint for α_m	Upper endpoint for α_m
5	1	2.0000000000	2.4580325859
10	2	2.7001330054	2.9828910906
20	3	3.2335773659	3.3677623965
50	5	3.6481708456	3.6845734707
100	7	3.8074760932	3.8201232724

to provide simple, independently checkable examples of the new all- m scalar corridor without computing a large characteristic polynomial. For $m \leq 20$, Part I's rational Perron certificates (section 8) are much sharper.

17.3 Floating-point component and large-bound calculations

The separate program `code/02-centered-remainder-compute.py` evaluates sparse component matrices and scalar roots. Its component values reproduce the known small-bound pattern (compare table 2) and illustrate the new upper bound. For larger m , no matrix eigensolver is

Table 8: Numerical growth rates; displayed digits are not interval certificates. The source file `data/02-centered-remainder-component_numerics.csv` also has a row for $m = 30$.

m	$\lambda_U(m)$	$\lambda_V(m)$	α_m	β_m
2	1.465571232	1.000000000	1.465571232	1.618033989
3	1.826515772	1.691402554	1.826515772	2.000000000
4	2.100456128	2.091476748	2.100456128	2.263408655
5	2.312119493	2.351509000	2.351509000	2.458032586
10	2.902174743	2.952689154	2.952689154	2.982891091
20	3.333400105	3.356795518	3.356795518	3.367762396
50	3.675866778	3.681914231	3.681914231	3.684573471

needed to evaluate the scalar corridor. Normalized coefficients b_k are generated by

$$b_1 = \frac{1}{4}, \quad \frac{b_{k+1}}{b_k} = \frac{2k-1}{2(k+1)}.$$

Solving for $t = m \log(4x)$ avoids subtracting a tiny root displacement from $1/4$. The block coefficients are also normalized, using the consecutive first-deficiency ratio obtained from (21).

Table 9: Floating-point evaluations with $d = \lceil \log_2 m \rceil$. The mathematical endpoints are rigorous bounds; these nearest-rounded numerical evaluations are not themselves certified intervals. From `data/02-centered-remainder-scalar_numerics.csv` (63 values of m).

m	d	$s_{m,d}^{-1}$	β_m	Corridor width
10^2	7	3.807476093297	3.820123272395	$1.265 \cdot 10^{-2}$
10^3	10	3.975171751563	3.975423985146	$2.522 \cdot 10^{-4}$
10^4	14	3.996950887245	3.996955062448	$4.175 \cdot 10^{-6}$
10^5	17	3.999640264705	3.999640325443	$6.074 \cdot 10^{-8}$
10^6	20	3.999958715503	3.999958716333	$8.301 \cdot 10^{-10}$

At $m = 10^6$, the two numerical bounds for E_m are approximately 3.149478 and 3.150308, although $2 \log \pi \approx 2.289460$. This is compatible with the theorem: the leading correction to the constant is $(8z - 12)/\log m$, which decays very slowly. A close upper and lower corridor is not evidence that its center is already near its limit.

Remark 17.1 (Merge note: the numerics are not evidence for the value of the limit). The corridor in figure 3 sits about 0.86 above $2 \log \pi$ at $m = 10^6$, and its lower endpoint is not even monotone in the computed range: it rises to about 3.168 near $m = 73,920$ and then decreases slowly. At $m = 10^6$ the first correction term $(8z - 12)/\log m$ of (79) is about 0.983, and the three displayed corrections together give $2 \log \pi + 0.983 - 0.104 - 0.005 \approx 3.164$, within 0.015 of the corridor. That agreement shows that the computations are consistent with the inverse-logarithmic corrections. It is *not* evidence for the value $2 \log \pi$ of the limit: no computation in the range $m \leq 10^6$ distinguishes that constant from nearby ones, and a slowly drifting corridor centred near 3.15 would look the same whatever the limit were. The value $2 \log \pi$ rests on the proofs of sections 12 to 15 alone. The decimals quoted in this remark were computed for the merge from `data/02-centered-remainder-scalar_numerics.csv`.

The computations do not estimate an exact leading constant for the scalar defect, certify component ordering beyond the tested values, or validate an infinite asymptotic expansion by numerical fitting.

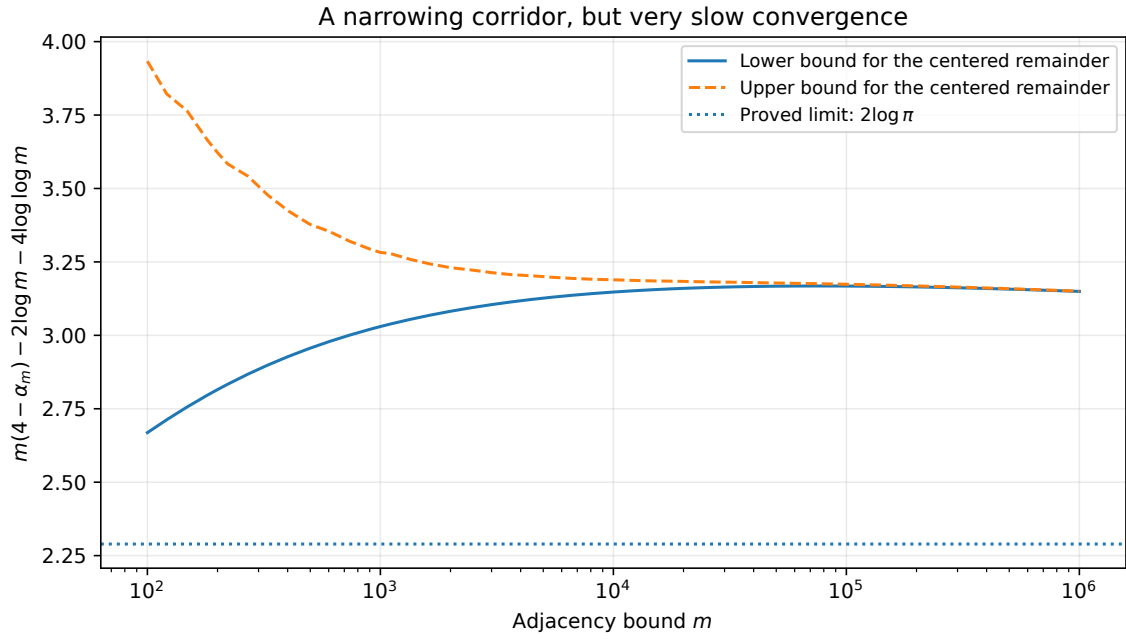


Figure 3: Numerical evaluations of the lower and upper bounds for E_m , not direct evaluations of the unknown large- m component maximum. The dotted line is the proved limiting constant. The asymptotic bounds need not be monotone, and the visible range has not reached the limit. (Manuscript 04's figure; compare figure 2, whose bounds are Part I's.)

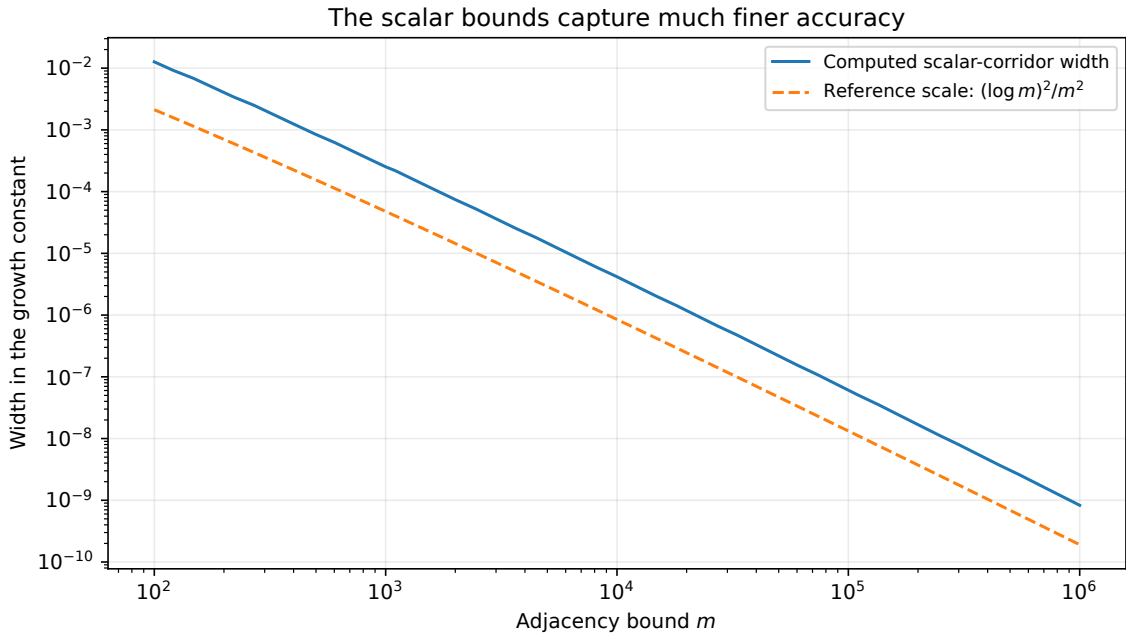


Figure 4: The numerical width $\beta_m - s_{m,d}^{-1}$ and the comparison scale $(\log m)^2/m^2$. The dashed curve has coefficient one solely for scale comparison; no claim is made that it is a finite- m upper bound or the optimal asymptotic equivalent.

18 A complete small example and a useful non-implication

At $m = 2$, order the U states by $p = 0, 1$, and the V states as $(1, 0), (0, 1)$. Then

$$W_{U_2}(x) = \begin{pmatrix} 0 & x \\ x^2 & x \end{pmatrix}, \quad W_{V_2}(x) = \begin{pmatrix} 0 & x \\ x & 0 \end{pmatrix}.$$

Thus $\lambda_U(2)$ is the positive root of $\lambda^3 - \lambda^2 - 1 = 0$, while $\lambda_V(2) = 1$ (the matrix W_{U_2} also appears in the proof of theorem 1.1). The scalar polynomial is $K_2(x) = x + x^2$, and $\beta_2 = (1 + \sqrt{5})/2$.

The last-threshold vector is $v = (1, 1 + x)^\top$. At the root $K_2(r_2) = 1$,

$$W_{V_2}(r_2)v = \begin{pmatrix} r_2 + r_2^2 \\ r_2 \end{pmatrix} = \begin{pmatrix} 1 \\ r_2 \end{pmatrix} <_{\text{some}} \begin{pmatrix} 1 \\ 1 + r_2 \end{pmatrix} = v,$$

where $<_{\text{some}}$ means coordinatewise inequality with at least one strict coordinate. The separate reset comparator of lemma 12.1 is

$$N(x) = \begin{pmatrix} 0 & x \\ x & x \end{pmatrix}, \quad \det(I - N(x)) = 1 - x - x^2.$$

This differs from $\det(I - W_{V_2}(x)) = 1 - x^2$, illustrating exactly why a spectral comparison is not an equality of state systems.

There is also an immediate counterexample to the tempting coefficientwise interpretation. For length three and $m = 2$, every 132-avoider meets the adjacency bound, so $[x^3]A_2(x) = C_3 = 5$ (table 1). But

$$[x^3] \frac{1}{1 - K_2(x)} = [x^3] \frac{1}{1 - x - x^2} = 3.$$

The upper bound on exponential growth is valid even though this proposed coefficientwise majorant is false. This example is a useful diagnostic when transporting the renewal argument to another combinatorial class.

19 Further research questions and proposed approaches

The exact constant is no longer the first obstruction. The quantitative corridor changes which questions are mathematically distinguishable and which tools are likely to be useful. The second question below is Part I's first open question (section 9), now constrained by corollary 15.4; the eighth continues Part I's remark on minimal recurrence orders and denominators.

19.1 The first term beyond scalar universality

Determine the true order and leading behavior of $\beta_m - \alpha_m$. Our upper bound is $O((\log m)^2/m^2)$, obtained by discarding a logarithmic number of terminal block lengths and controlling a deficiency tail uniformly. Those losses need not be optimal. A sharper approach would keep the compatibility between a block length and the next first deficiency rather than impose a single worst-case cutoff. A small matrix of deficiency classes, with a positive-vector comparison analogous to lemma 12.2, is a concrete candidate. No asymptotic equivalent for this defect is asserted here.

19.2 The eventual ordering of the two components

Resolve whether $\lambda_V(m) > \lambda_U(m)$ for every $m \geq 5$, the ordering suggested by finite computations and related to Problem P3 in [2]. By corollary 15.4, a proof cannot be obtained merely by comparing their shared inverse-logarithmic expansion with the scalar root. It must detect finer endpoint losses or construct a comparison preserving more information than the present supersolution. One possibility is to compare first-return operators after isolating the large-threshold states, where long blocks are compatible with nearly all first entries.

19.3 The geometry of near-critical Perron vectors

Describe the positive eigenvectors of W_{U_m} and W_{V_m} at their critical roots as $m \rightarrow \infty$. The scalar tilted mass is concentrated in a layer of width about m/t_m near the maximal block length, while the chosen cutoff is only of order $\log m$. A useful question is whether endpoint coordinates admit a scaling limit connecting these two regimes. Such a limit could explain the loss of the scalar upper bound and supply an analytic route to the sign of the component gap. The present test vector is a supersolution, not a proved approximation to the true eigenvector.

19.4 Poles, amplitudes, and a uniform spectral gap

For each fixed m , determine when the dominant pole of A_m is simple and obtain a usable expression for the leading amplitude. An equality of component rates could interact with paths between components and alter pole order. Our argument identifies only their maximal exponential rate. A uniform analysis of subdominant singularities would be needed to turn fixed- m spectral information into estimates with useful error constants as m grows. This is distinct from the scalar root's own simple zero, which follows immediately from $K'_m(r_m) > 0$.

19.5 Joint growth of length and adjacency bound

Study $a_n^{(m(n))}/C_n$ when $m = m(n)$ grows with n . The fixed- m limit used here is not automatically uniform, so substituting $m(n)$ into α_m^n is not justified. Candidate regimes should compare the exponential penalty $n \log(4/\alpha_m)$ with the finite-state mixing scale and the Catalan polynomial factor. A satisfactory result would establish an explicit range of uniformity before proposing a transition law. The identity $a_n^{(m)} = C_n$ when $m \geq n - 1$ provides an exact boundary condition for such a theory.

19.6 Renewal comparisons for other constrained tree classes

The reset identity lemma 12.1 is not specifically Catalan. It applies whenever a monotone threshold is lowered by weighted steps and restored by a reset. Identify other pattern-avoidance or tree-enumeration problems where the associated renewal sequence is explicitly controlled. If a truncated scalar coefficient law has a summable power tail, an analogous endpoint tilt should be investigated. The necessary research task is to prove a matching combinatorial lower construction; a scalar majorant alone does not make its asymptotic constant universal.

19.7 Weighted statistics and multivariate perturbations

Refine the endpoint recurrence with variables recording descents, records, or block statistics, in the direction of Problem P4 of [2]. The central question is whether a positive last-threshold witness persists under a parameter perturbation. A successful extension might permit derivative bounds for a weighted growth rate and limit laws at fixed m . Positivity and irreducibility must be checked in the perturbed system; formal differentiation of an unproved multivariate analogue of K_m would be insufficient.

19.8 State minimization versus a spectral majorant

Investigate whether the renewal witness can guide an exact reduction of the endpoint system without confusing domination with equality. Problems P1 and P2 in [2] concern issues such as state complexity and rational denominators. Our m -state reset comparator has determinant $1 - K_m$, but it forgets first-endpoint compatibility. One concrete goal is a characterization of the lost information as a correction operator of small rank or controlled support. An exact operator identity would be a substantive advance beyond the present scalar comparison.

19.9 Certified computation and formalization

Build an adaptive exact or interval algorithm choosing d to meet a requested error tolerance, using K_m and $K_{m,d}$ directly. This would avoid large component determinants and would certify large- m growth intervals with clear cost accounting. For formalization, the finite renewal polynomial identity, the spectral supersolution argument, the skew-cut injection, and the uniform deficiency-tail inequality are natural first targets. The analytic tail splitting and all-orders inversion can then be formalized as separate lemmas with explicit uniform quantifiers. None of this formalization has been started.

20 Proof boundaries, reproducibility and provenance

20.1 What the manuscript has established

The proof route can be read as a dependency chain:

endpoint system $\implies \alpha_m < \beta_m \implies$ sharp scalar corridor $\implies$ exact constant and all orders.

The lower arrow uses both recoverable skew blocks and the uniform first-entry tail; the analytic arrow uses the finite endpoint expansion with its stated remainder. No repository assertion about strict monotonicity is needed for the new theorems: [theorem 1.1](#) is not used. The finite-state growth interface is proved in Part I and was rederived in the manuscript, so it is also not being accepted solely on the authority of a software claim.

Table 10: Trust and scope ledger of Part II.

Statement or artifact	Validation status
All- m upper bound and asymptotic theorems	Written mathematical proofs here
Finite row, block, and tail tests	Executed exact integer/rational checks
First four inverse-log polynomials	Executed exact symbolic checks
Large- m numerical figures and component table	Floating-point illustrations only
New Lean or Rocq theorem declarations	None supplied or claimed
External peer review or exhaustive novelty verification	Not performed
Component ordering, optimal scalar defect, joint limits	Open directions (section 19)

20.2 Reproducing Part II's computations

Part II's files are

code/02-centered-remainder-model.py	Exact Catalan, endpoint, and block objects
code/02-centered-remainder-verify.py	Standard-library exact verification
code/02-centered-remainder-verify_expansion.py	Exact symbolic coefficient check
code/02-centered-remainder-compute.py	Floating-point numerical illustrations
code/02-centered-remainder-plot_results.py	The two figures from the CSV data
data/02-centered-remainder-*	Check reports, rational certificates, CSVs, environment, requirements, records of the delivered PDF
figures/02-centered-remainder-*.pdf	The two figures of this part

The scripts are shipped byte-identical to the delivery and still use the delivered names. They write

data/verification_report.json	data/rational_certificates.json
data/expansion_verification.json	data/scalar_numerics.csv
data/component_numerics.csv	figures/remainder_corridor.{pdf,png}

figures/corridor_width.{pdf,png}

and the verifier and the numerical program execute from `model import`. Run in place, that import would load Part I's different code/`model.py` and fail with `ImportError`, and the outputs would land beside Part I's unprefixed files. The report's README gives a recipe that restores the delivered layout in a scratch copy. There, the exact checks need Python 3.10 or newer and no third-party packages (python code/`verify.py`; do not use `-O`, the verifier uses assertions); the symbolic check needs SymPy; the numerical programs need NumPy, SciPy and, for the plots, Matplotlib (versions in `data/02-centered-remainder-environment.json`). A POSIX `OPENBLAS_NUM_THREADS=1` prefix on the numerical command is optional; it limits numerical-library threading rather than changing any mathematical assertion. The retained figures suffice to rebuild the article without rerunning any computation.

20.3 Points checked in the written argument

The sharp upper bound is not obtained by dropping a legitimate append term: that term is incorporated by the last-threshold vector. The first-entry tail estimate is uniform in both parameters, which permits a growing cutoff. The conversion from a polynomial value deficit to a root displacement uses a proved derivative lower bound, not a heuristic linearization. The order of inequalities is preserved when reciprocal roots become growth bounds and when growth bounds become deficit bounds. Finally, the factorially divergent formal series is used only to define finite coefficient calculations, while each analytic statement has its own fixed-order remainder.

These checks identify the parts of the proof most worth independent scrutiny. They do not replace that scrutiny, but they make its targets explicit and keep finite experiments distinct from universal arguments.

20.4 Repository provenance and novelty boundary

The manuscript inspected Part I at the repository-relative path

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/article.tex
```

at the ProveIt commit

```
6c9175a54bcaad3bfc2d416257d2b7d3dff2c1e0
```

where its Git blob identifier was

```
5ec75ee48df94b6d12804a26ba109c08564df5c7
```

Its section “What is resolved, and what remains” (section 9) explicitly asks whether E_m converges and describes the unresolved constant window; the opening theorem statements specify the window in (3). The finite endpoint system and lower block construction used here are attributed to their prior sources, with independent proofs supplied in the manuscript.

The manuscript's primary-paper check used the arXiv version-one records and HTML sources for [1, 2], accessed on September 28, 2026. The inspected materials did not contain the last-threshold supersolution, the sharp scalar corridor, or the exact constant proved here. This is a bounded provenance statement. It does not certify the absence of an independent unpublished argument, a later repository change, or an uninspected paper.

20.5 Where the merge had to choose

- *Placement.* Part II follows Part I's last main section and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Part I changed.
- *Duplicates.* The endpoint recurrence, the component classification and spectral interface (the manuscript's Section 2 and Appendix A), the first-entry lemma and the injective skew-block construction are printed once, in Part I, and credited in sections 11 and 13. The manuscript's different argument for $\lambda_U(m) > 1$ is kept as a second proof. The second proof of corollary 14.2 and remarks 12.3, 13.3 and 17.1 are the merge's.
- *Notation.* $L \rightarrow \Lambda$, $P_j \rightarrow \mathcal{P}_j$, $B_N \rightarrow \mathcal{B}_N$, $T_m \rightarrow \Delta_m$, $C, c \rightarrow C_*, c_*$ (absolute constants), $W_U, W_V \rightarrow W_{U_m}, W_{V_m}$, and $H = \log m$ written out (table 4). No normalization changed.
- *References to Part I.* The manuscript cites Part I as a repository report; here those citations are internal cross-references, and the manuscript's bibliography entries for [1, 2] are Part I's.
- *Reproduction.* The manuscript's archive listing and commands are replaced by the shipped names and the copy recipe (section 20.2); its `article.tex`, `article.pdf`, `README.md` and the PNG versions of its figures are not shipped.
- *Figures.* In the write phase the manuscript's two figure PDFs were copied byte-identically into `figures/`, under their delivered names with the prefix `02-centered-remainder-` (figures 3 and 4).
- *Stale sentences in Part I.* The abstract, theorem 1.3, the remark after theorem 1.3's proof, section 9 and section B carry dated pointers to Part II; their original wording is kept.

A Exact small-bound algebra

For $m = 3$, the certified generating function is $A_3(x) = P_3(x)/Q_3(x)$, where

$$\begin{aligned} P_3(x) &= 1 - x - x^2 + x^3 + 4x^4 - 4x^6 - 3x^7 - 3x^8 \\ &\quad - 5x^9 - 3x^{10} + 2x^{11} + 2x^{12}, \\ Q_3(x) &= (x - 1)(x + 1)(x^5 - x^4 - x^3 + x^2 - 2x + 1) \\ &\quad \cdot (x^6 + x^5 + x^4 + 2x^3 + x^2 - 1). \end{aligned}$$

For $m = 4$, the denominator factors as

$$\begin{aligned} Q_4(x) &= (x - 1) D_U(x) D_V(x), \\ D_U(x) &= x^{10} + x^8 + x^7 - 6x^5 - x^4 - x^3 - x^2 - x + 1, \\ D_V(x) &= x^{12} - x^{11} - x^{10} - x^9 + 5x^8 + 4x^7 + 4x^6 \\ &\quad + 4x^5 + 5x^4 + 3x^3 + x^2 - 1. \end{aligned}$$

Both displayed full denominators have constant term one. The numerator of A_4 , listed from constant term upwards, has coefficient vector

$$(1, -1, -1, -1, 2, 9, 5, -3, -14, -10, -13, -23, \\ -31, -11, 21, 32, 21, 1, -3, -5, -3, -3).$$

It has degree 21 and Q_4 has degree 23. Thus the usual coefficient recurrence from $Q_4 A_4 = P_4$ is valid for all $n \geq 23$ when written using 23 preceding terms. The complete ascending coefficient arrays, full state certificates, and exact checks are in the JSON files.

B Reproduction and certificate formats

The archive is arranged as follows:

article.tex, article.pdf	Main editable source and report
README.md, requirements.txt	Reproduction instructions
code/model.py	Exact endpoint model and combinatorics
code/compute.py	Numerical proposals and exact symbolic solver
code/verify.py	Standard-library independent exact verifier
code/plot_results.py	Recreates the two separate figures
data/sequences.csv	$n = 0..200$, $m = 1..12$
data/growth_constants.csv	Components for $m = 2..20$
data/perron_certificates.json	Rational bounds and positive integer vectors
data/symbolic_certificate_m*.json	Polynomial state certificates, $m = 1..4$
data/generating_functions.json	Reduced rational functions, $m = 1..4$
data/large_m_scalar_bounds.csv	Scalar evaluations through $m = 1000000$

[Added 28 September 2026, batch 38: Part II's code, data and figures carry the prefix 02-centered-remainder- and are listed in [section 20.2](#).]

The primary reproducibility command needs no third-party Python packages:

```
python code/verify.py
```

To regenerate numerical and symbolic proposals, install the packages in requirements.txt and run

```
python code/compute.py --max-m 20 --symbolic-max 4 --large-scalars
python code/verify.py
python code/plot_results.py
```

The generator deliberately raises an exception if it cannot convert a numerical proposal to a valid exact certificate. A floating-point result alone is never accepted as a proof of a spectral inequality.

To build the report from the supplied figures:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The third pass settles page references and the table of contents if needed. The TeX source uses standard packages from a current full TeX Live installation; no font files are distributed.

The exact recurrence uses $(m + 1)^2$ endpoint states and at most $m + 1$ terms per state. Computing all coefficients through length N therefore uses $O(Nm^3)$ integer multiply-add operations, after coefficient precomputation, and $O(Nm^2)$ stored integers in the supplied straightforward implementation. These are arithmetic-operation counts, not bit-complexity bounds. Exact values have growing bit length. A rolling buffer can reduce storage when the complete coefficient table is not needed.

Perron certificates store integers p, D for each radius endpoint p/D , and a positive integer vector in the state order generated by the verifier. Symbolic certificates store all polynomial coefficients in ascending degree. The constant term of each common denominator is checked to be one. The certificate identities prove infinitely many coefficients because the formal system has a unique solution; the finite coefficient checks are additional consistency tests.

C Source status and proof audit

The literature check was performed on September 19, 2026. The inspected primary items are the arXiv v1 versions listed below, including Nadler’s Conjecture 2 and Mayama–Akita’s Theorem 4.14 and Problem P3. The latter paper’s displayed numerical table and statement that its component comparison was unproved were also inspected in the PDF. Targeted searches used the paper titles and identifiers together with strict monotonicity, growth constants, and asymptotic refinements. No later solution was located in that search. This establishes the specific source-based status used here; it is not an assertion that every possible publication has been searched.

The logical dependencies of the new claims are particularly short. The strict-growth theorem uses the exact endpoint recurrence, the two irreducible components, their root characterization, the proper-principal Perron lemma, and the explicit shift of thresholds. This report includes proofs of those prerequisites. The asymptotic theorem instead uses the maximum-decomposition majorant, the injective skew-block construction, the first-entry Catalan formula, and the truncated-tail lemma. In particular, no numerical fit, unproved component-ordering pattern, or formal passage from finite checks to all m is used.

The work has not been formalized in a proof assistant. The integer and polynomial certificates are executable verification of the stated finite algebraic claims; they should not be described as a formal verification of the analytic limiting argument or of the all- m theorem.

References

- [1] N. Nadler, *On 132-Avoiding Permutations with an Adjacency Constraint*, arXiv:2604.22135v1, submitted April 24, 2026. <https://arxiv.org/abs/2604.22135>. Conjecture 2 is in Section 5.2.
- [2] T. Mayama and D. Akita, *Finite-state enumeration of adjacency-constrained 132-avoiding permutations*, arXiv:2605.23519v1, submitted May 22, 2026. <https://arxiv.org/abs/2605.23519>. Relevant locations: Theorem 2.4, Proposition 2.7, Theorems 4.12 and 4.14, and Problem P3 in Section 5.